

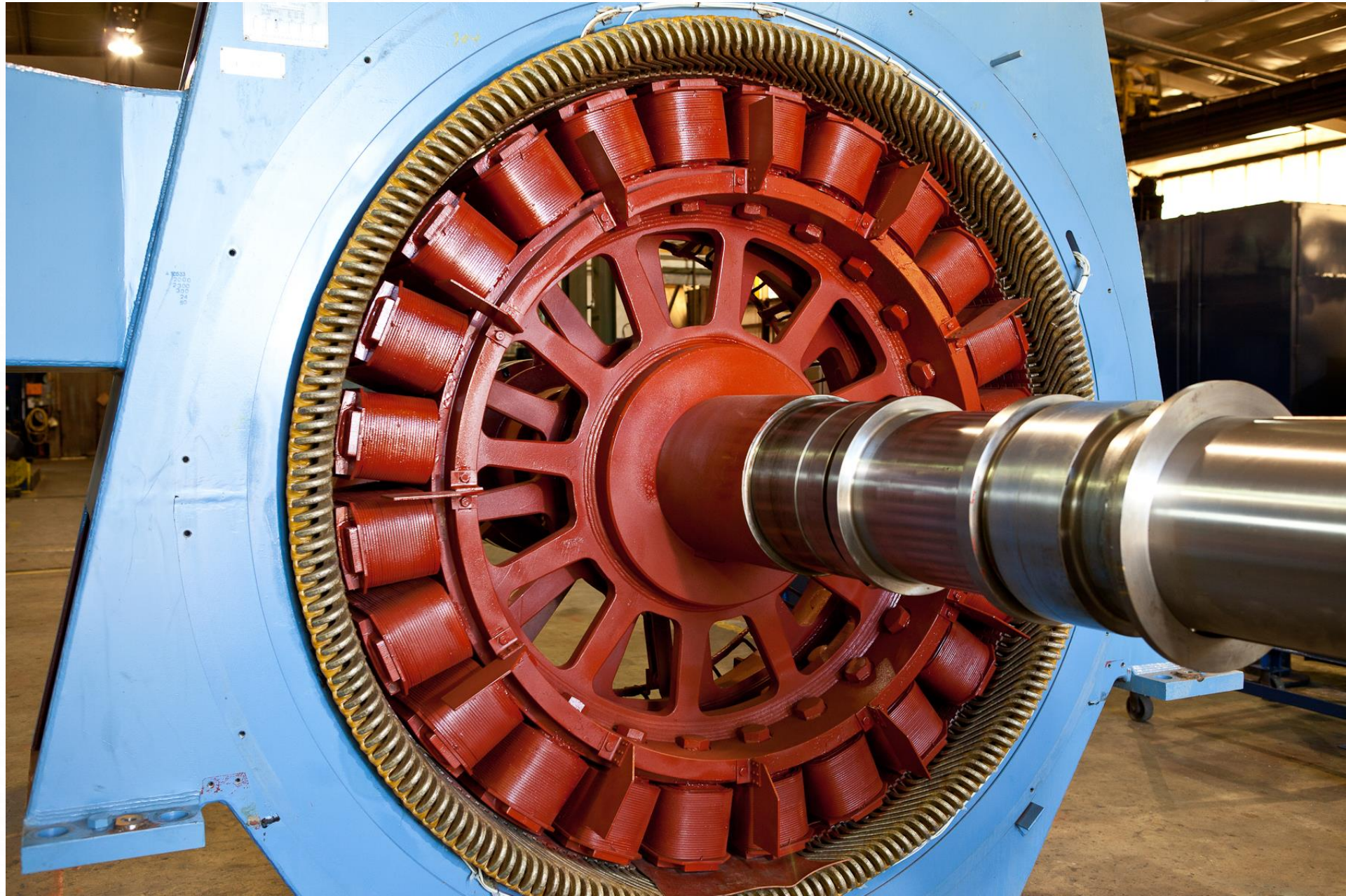
# SYNCHRONOUS MOTOR



The following slides contain copyright materials under fair use policy for educational purpose only. Any commercial use will be subjected to the violation of copyright law.



# CONSTRUCTION (13.1)





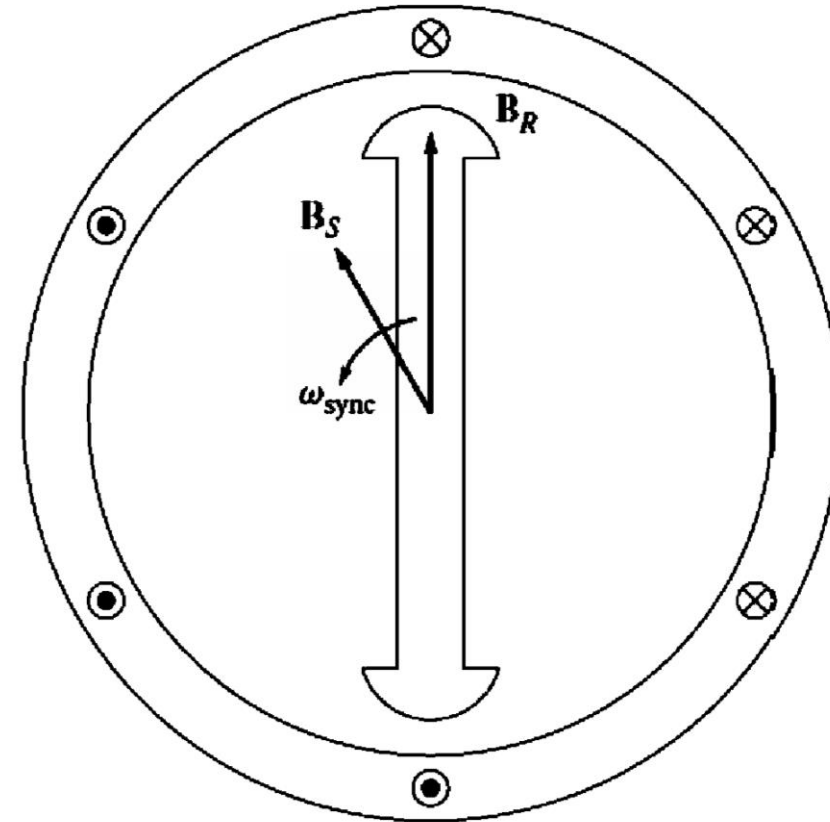
# SOME FACTS ABOUT SM

- (i) A synchronous motor runs at synchronous speed or not at all. Its speed is constant (synchronous speed) at all loads. The only way to change its speed is to alter the supply frequency ( $N_s = 120 f/P$ ).
- (ii) The outstanding characteristic of a synchronous motor is that it can be made to operate over a wide range of power factors (lagging, unity or leading) by adjustment of its field excitation. Therefore, a synchronous motor can be made to carry the mechanical load at constant speed and at the same time improve the power factor of the system.
- (iii) Synchronous motors are generally of the salient pole type.
- (iv) A synchronous motor is not self-starting and an auxiliary means has to be used for starting it. We use either induction motor principle or a separate starting motor for this purpose. If the latter method is used, the machine must be run up to synchronous speed and synchronized as an alternator.

# BASIC PRINCIPLES OF MOTOR OPERATION (6.1)

To understand the basic concept of a synchronous motor, look at Figure 6-1, which shows a two-pole synchronous motor. The field current  $IF$  of the motor produces a steady-state magnetic field  $B_R$ . A three-phase set of voltages is applied to the stator of the machine, which produces a three-phase current flow in the windings.

As was shown earlier a three-phase set of currents in an armature winding produces a uniform rotating magnetic field  $B_S$ . Therefore, there are two magnetic fields present in the machine, and the **rotor field will tend to line up with the stator field**, just as two bar magnets will tend to line up if placed near each other. Since the stator magnetic field is rotating, the rotor magnetic field (and the rotor itself) will constantly try to catch up. The larger the angle between the two magnetic fields (up to a certain maximum), the greater the torque on the rotor of the machine. The basic principle of synchronous motor operation is that the rotor "chases" the rotating stator magnetic field around in a circle, never quite catching up with it.



$$\tau_{ind} = k B_R \times B_S \\ = \text{counterclockwise}$$



# THE ROTATING MAGNETIC FIELD (4.2)

$$i_{aa'}(t) = I_M \sin \omega t \quad \text{A}$$

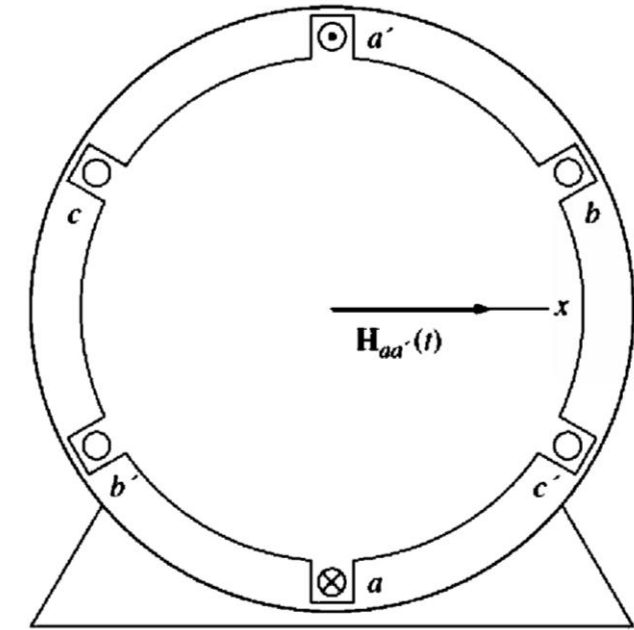
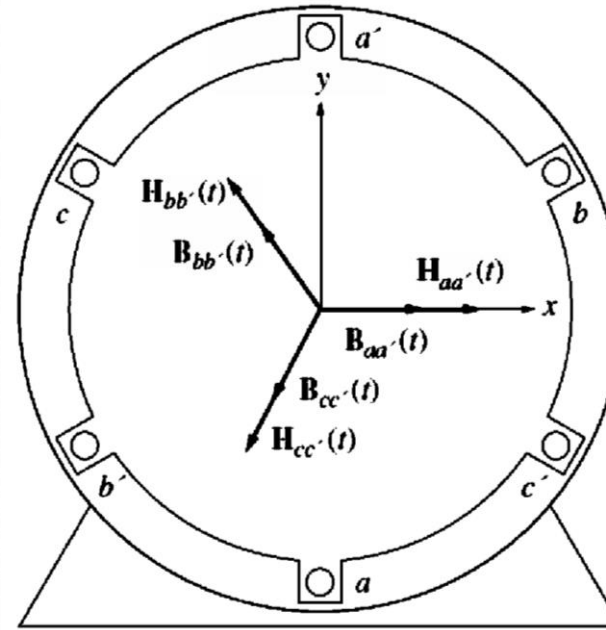
$$i_{bb'}(t) = I_M \sin (\omega t - 120^\circ) \quad \text{A}$$

$$i_{cc'}(t) = I_M \sin (\omega t - 240^\circ) \quad \text{A}$$

$$H_{aa'}(t) = H_M \sin \omega t \angle 0^\circ \quad \text{A} \cdot \text{turns} / \text{m}$$

$$H_{bb'}(t) = H_M \sin (\omega t - 120^\circ) \angle 120^\circ \quad \text{A} \cdot \text{turns} / \text{m}$$

$$H_{cc'}(t) = H_M \sin (\omega t - 240^\circ) \angle 240^\circ \quad \text{A} \cdot \text{turns} / \text{m}$$



The flux densities resulting from these magnetic field intensities are given by Equation (1-21):

$$\mathbf{B} = \mu \mathbf{H} \quad (1-21)$$

They are

$$\mathbf{B}_{aa'}(t) = B_M \sin \omega t \angle 0^\circ \quad \text{T} \quad (4-23a)$$

$$\mathbf{B}_{bb'}(t) = B_M \sin (\omega t - 120^\circ) \angle 120^\circ \quad \text{T} \quad (4-23b)$$

$$\mathbf{B}_{cc'}(t) = B_M \sin (\omega t - 240^\circ) \angle 240^\circ \quad \text{T} \quad (4-23c)$$

For example, at time  $\omega t = 0^\circ$ , the magnetic field from coil  $aa'$  will be

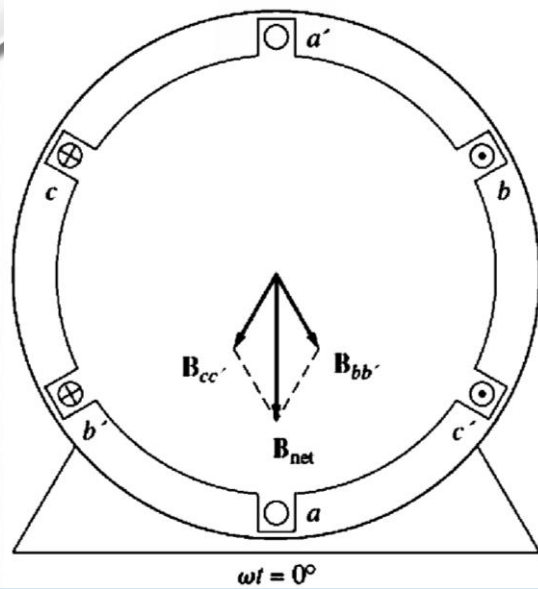
$$\mathbf{B}_{aa'} = 0 \quad (4-24a)$$

The magnetic field from coil  $bb'$  will be

$$\mathbf{B}_{bb'} = B_M \sin (-120^\circ) \angle 120^\circ \quad (4-24b)$$

and the magnetic field from coil  $cc'$  will be

$$\mathbf{B}_{cc'} = B_M \sin (-240^\circ) \angle 240^\circ \quad (4-24c)$$



The total magnetic field from all three coils added together will be

$$\begin{aligned} \mathbf{B}_{\text{net}} &= \mathbf{B}_{aa'} + \mathbf{B}_{bb'} + \mathbf{B}_{cc'} \\ &= 0 + \left(-\frac{\sqrt{3}}{2} B_M\right) \angle 120^\circ + \left(\frac{\sqrt{3}}{2} B_M\right) \angle 240^\circ \\ &= 1.5 B_M \angle -90^\circ \end{aligned}$$

As another example, look at the magnetic field at time  $\omega t = 90^\circ$ . At that time, the currents are

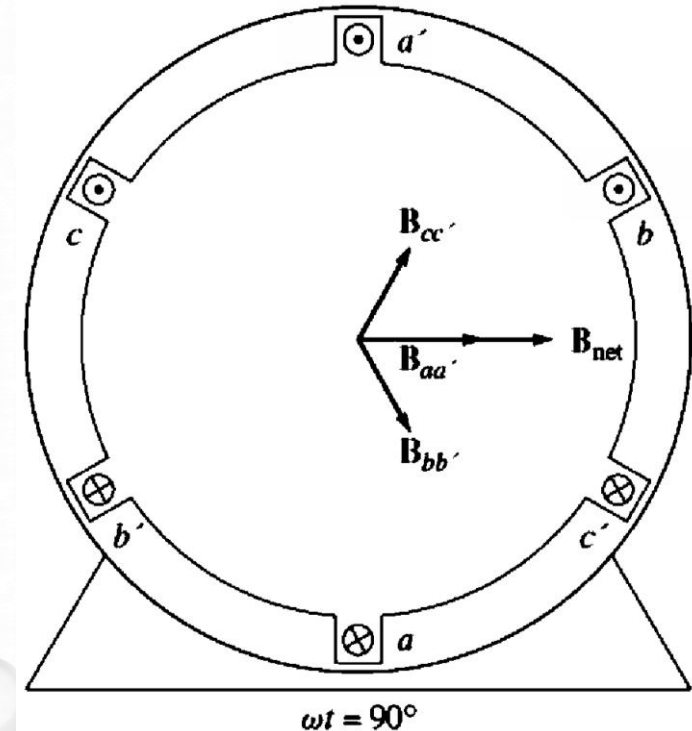
$$\begin{aligned} i_{aa'} &= I_M \sin 90^\circ & \text{A} \\ i_{bb'} &= I_M \sin (-30^\circ) & \text{A} \\ i_{cc'} &= I_M \sin (-150^\circ) & \text{A} \end{aligned}$$

and the magnetic fields are

$$\begin{aligned} \mathbf{B}_{aa'} &= B_M \angle 0^\circ \\ \mathbf{B}_{bb'} &= -0.5 B_M \angle 120^\circ \\ \mathbf{B}_{cc'} &= -0.5 B_M \angle 240^\circ \end{aligned}$$

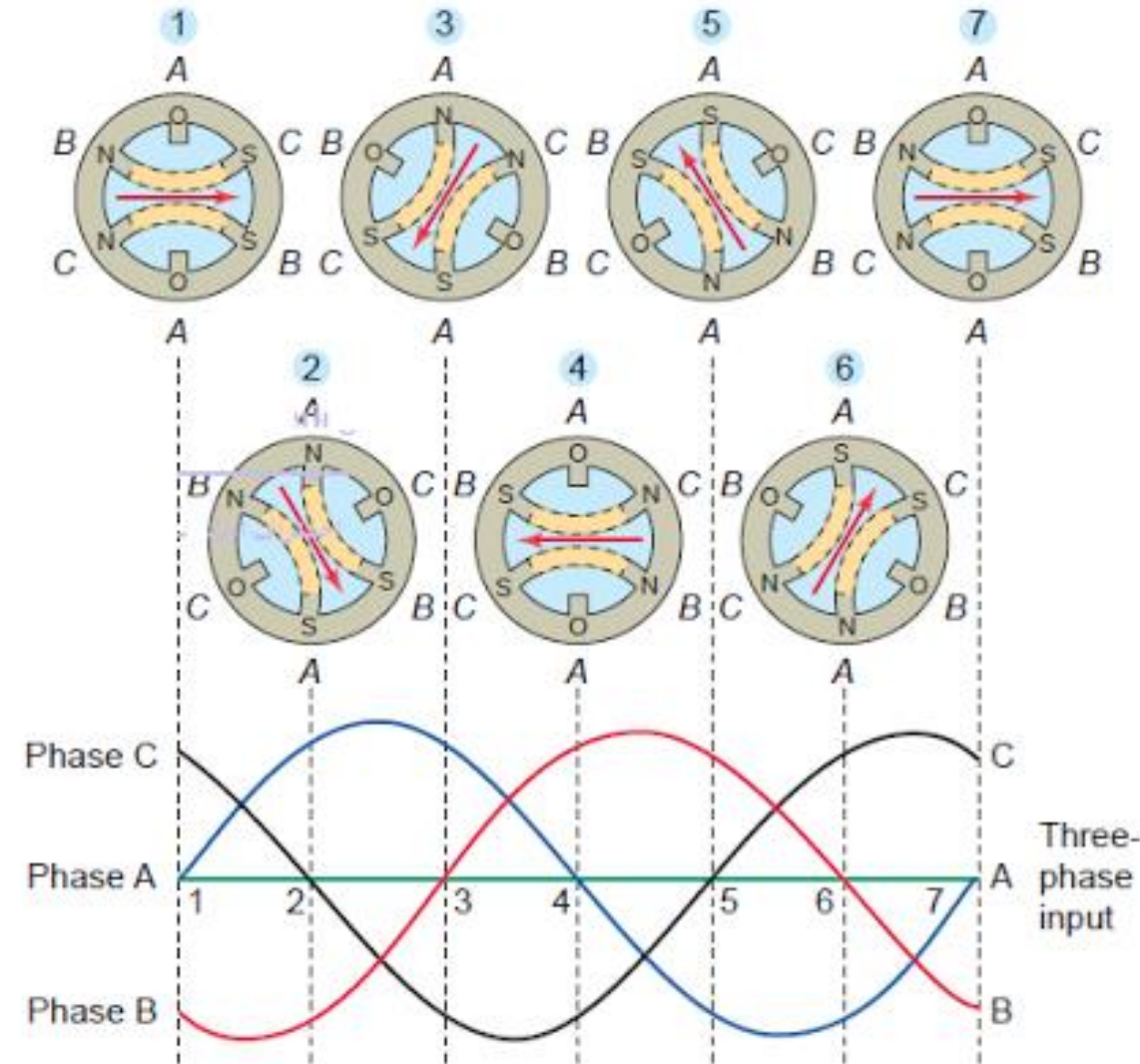
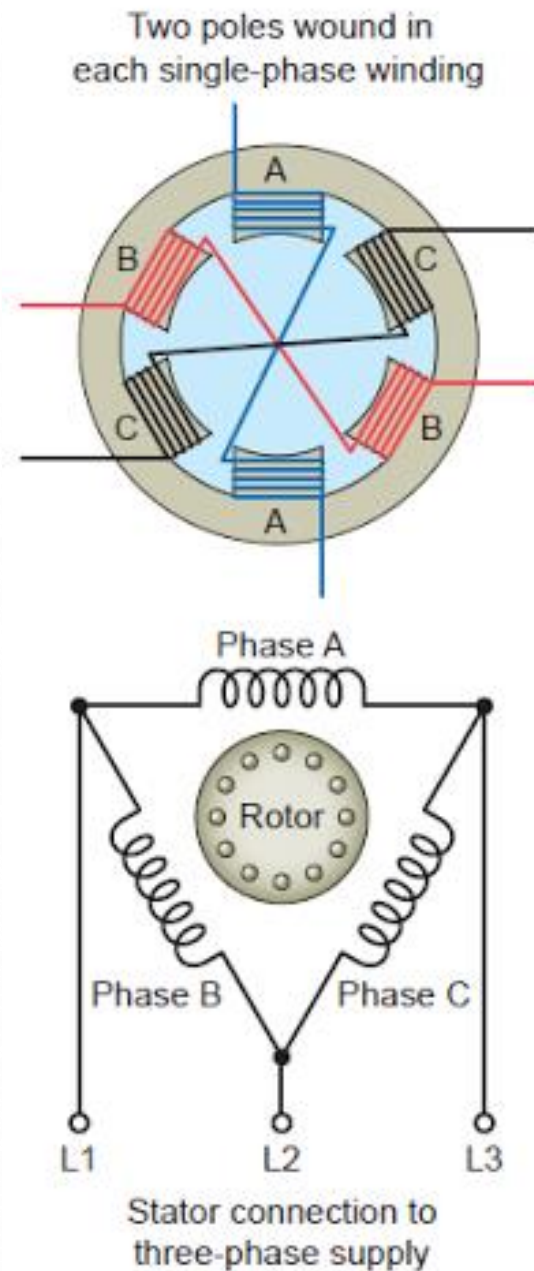
The resulting net magnetic field is

$$\begin{aligned} \mathbf{B}_{\text{net}} &= \mathbf{B}_{aa'} + \mathbf{B}_{bb'} + \mathbf{B}_{cc'} \\ &= B_M \angle 0^\circ + (-0.5 B_M) \angle 120^\circ + (-0.5 B_M) \angle 240^\circ \\ &= 1.5 B_M \angle 0^\circ \end{aligned}$$





Minimum 2 phases are needed to produce a rotating magnetic field maintaining that **the time angle displacement between the currents and space angle displacement between the winding axes.**



# SPEED OF SM

Also, noting that  $f_m = n_m/60$ , it is possible to relate the electrical frequency in hertz to the resulting mechanical speed of the magnetic fields in revolutions per minute. This relationship is

$$f_e = \frac{n_m P}{120} \quad (4-34)$$

$$\theta_e = \frac{P}{2} \theta_m$$

$$f_e = \frac{P}{2} f_m$$

$$\omega_e = \frac{P}{2} \omega_m$$

$$\text{Synchronous speed, } N_s = \frac{120f}{P}$$

where  $f$  = frequency of supply in Hz  
 $P$  = number of poles

Direction of rotating magnetic field: Clockwise or anti-clockwise depending on phase sequence. If the field rotates clockwise for 'abc' sequence, it will rotate anticlockwise for 'acb' sequence and vice-versa.



# INDUCED TORQUE IN AN AC MACHINE (4.5)

the rotor. The stator flux distribution in this machine is

$$B_s(\alpha) = B_s \sin \alpha \quad (4-51)$$

where  $B_s$  is the magnitude of the peak flux density;  $B_s(\alpha)$  is positive when the flux density vector points radially outward from the rotor surface to the stator surface.

The induced force on conductor 1 is

$$\begin{aligned} \mathbf{F} &= i(\mathbf{l} \times \mathbf{B}) \\ &= ilB_s \sin \alpha \quad \text{with direction as shown} \end{aligned} \quad (1-43)$$

The torque on the conductor is

$$\begin{aligned} \tau_{\text{ind},1} &= (\mathbf{r} \times \mathbf{F}) \\ &= rilB_s \sin \alpha \quad \text{counterclockwise} \end{aligned}$$

The induced force on conductor 2 is

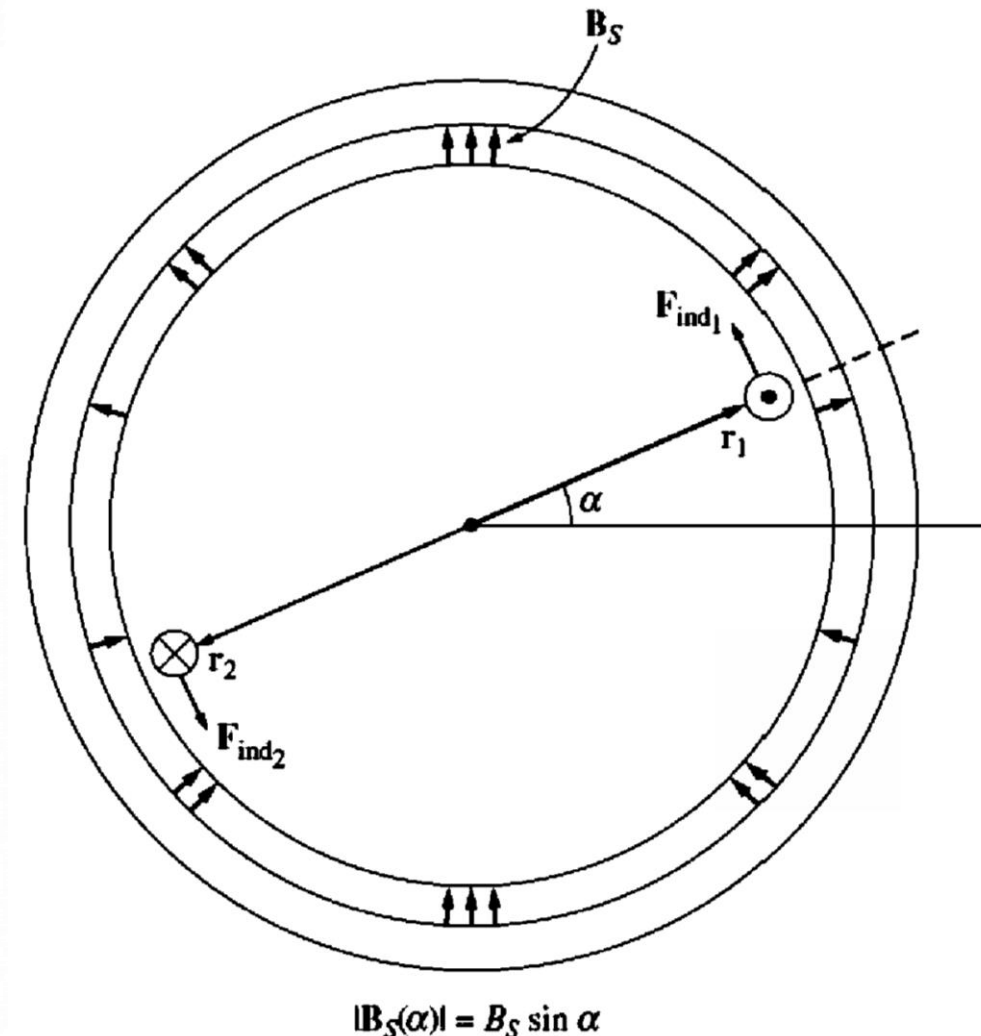
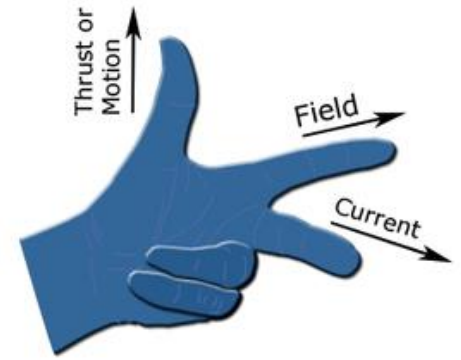
$$\begin{aligned} \mathbf{F} &= i(\mathbf{l} \times \mathbf{B}) \\ &= ilB_s \sin \alpha \quad \text{with direction as shown} \end{aligned} \quad (1-43)$$

The torque on the conductor is

$$\begin{aligned} \tau_{\text{ind},1} &= (\mathbf{r} \times \mathbf{F}) \\ &= rilB_s \sin \alpha \quad \text{counterclockwise} \end{aligned}$$

Therefore, the torque on the rotor loop is

$$\tau_{\text{ind}} = 2rilB_s \sin \alpha \quad \text{counterclockwise} \quad (4-52)$$



Equation (4-52) can be expressed in a more convenient form by examining Figure 4-18 and noting two facts:

1. The current  $i$  flowing in the rotor coil produces a magnetic field of its own. The direction of the peak of this magnetic field is given by the right-hand rule, and the magnitude of its magnetizing intensity  $\mathbf{H}_R$  is directly proportional to the current flowing in the rotor:

$$\mathbf{H}_R = Ci \quad (4-53)$$

where  $C$  is a constant of proportionality.

2. The angle between the peak of the stator flux density  $\mathbf{B}_S$  and the peak of the rotor magnetizing intensity  $\mathbf{H}_R$  is  $\gamma$ . Furthermore,

$$\gamma = 180^\circ - \alpha \quad (4-54)$$

$$\sin \gamma = \sin (180^\circ - \alpha) = \sin \alpha \quad (4-55)$$

By combining these two observations, the torque on the loop can be expressed as

$$\tau_{\text{ind}} = KH_R B_S \sin \alpha \quad \text{counterclockwise} \quad (4-56)$$

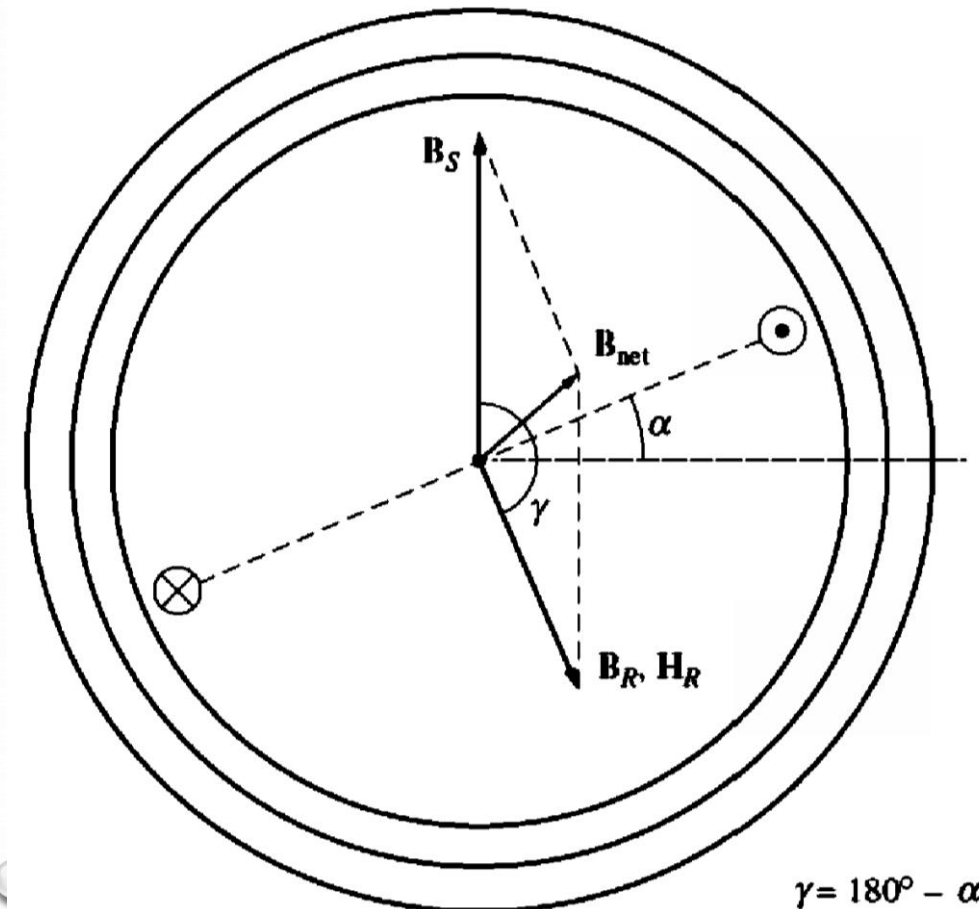
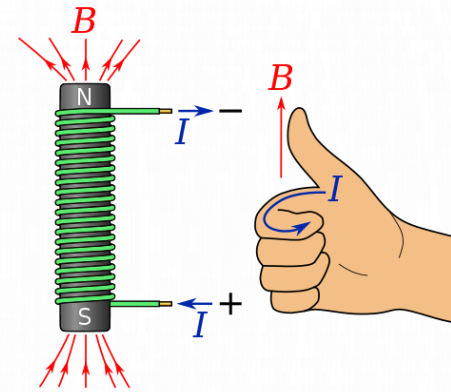
where  $K$  is a constant dependent on the construction of the machine. Note that both the magnitude and the direction of the torque can be expressed by the equation

$$\tau_{\text{ind}} = K \mathbf{H}_R \times \mathbf{B}_S \quad (4-57)$$

Finally, since  $B_R = \mu H_R$ , this equation can be reexpressed as

$$\tau_{\text{ind}} = k \mathbf{B}_R \times \mathbf{B}_S \quad (4-58)$$

where  $k = K/\mu$ . Note that in general  $k$  will not be constant, since the magnetic permeability  $\mu$  varies with the amount of magnetic saturation in the machine.





The net magnetic field in this machine is the vector sum of the rotor and stator fields (assuming no saturation):

$$\mathbf{B}_{\text{net}} = \mathbf{B}_R + \mathbf{B}_S \quad (4-59)$$

This fact can be used to produce an equivalent (and sometimes more useful) expression for the induced torque in the machine. From Equation (4-58)

$$\tau_{\text{ind}} = k \mathbf{B}_R \times \mathbf{B}_S \quad (4-58)$$

But from Equation (4-59),  $\mathbf{B}_S = \mathbf{B}_{\text{net}} - \mathbf{B}_R$ , so

$$\begin{aligned} \tau_{\text{ind}} &= k \mathbf{B}_R \times (\mathbf{B}_{\text{net}} - \mathbf{B}_R) \\ &= k(\mathbf{B}_R \times \mathbf{B}_{\text{net}}) - k(\mathbf{B}_R \times \mathbf{B}_R) \end{aligned}$$

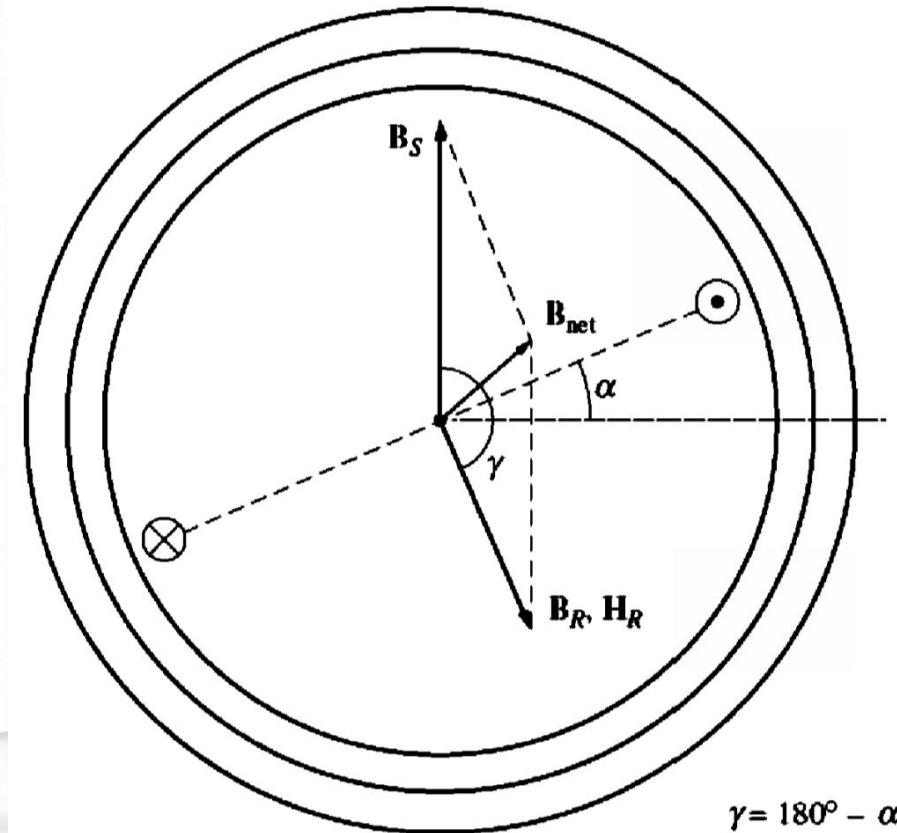
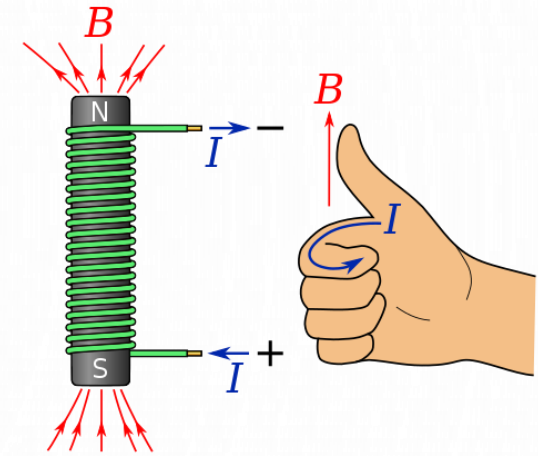
Since the cross product of any vector with itself is zero, this reduces to

$$\tau_{\text{ind}} = k \mathbf{B}_R \times \mathbf{B}_{\text{net}} \quad (4-60)$$

so the induced torque can also be expressed as a cross product of  $\mathbf{B}_R$  and  $\mathbf{B}_{\text{net}}$  with the same constant  $k$  as before. The magnitude of this expression is

$$\tau_{\text{ind}} = k B_R B_{\text{net}} \sin \delta \quad (4-61)$$

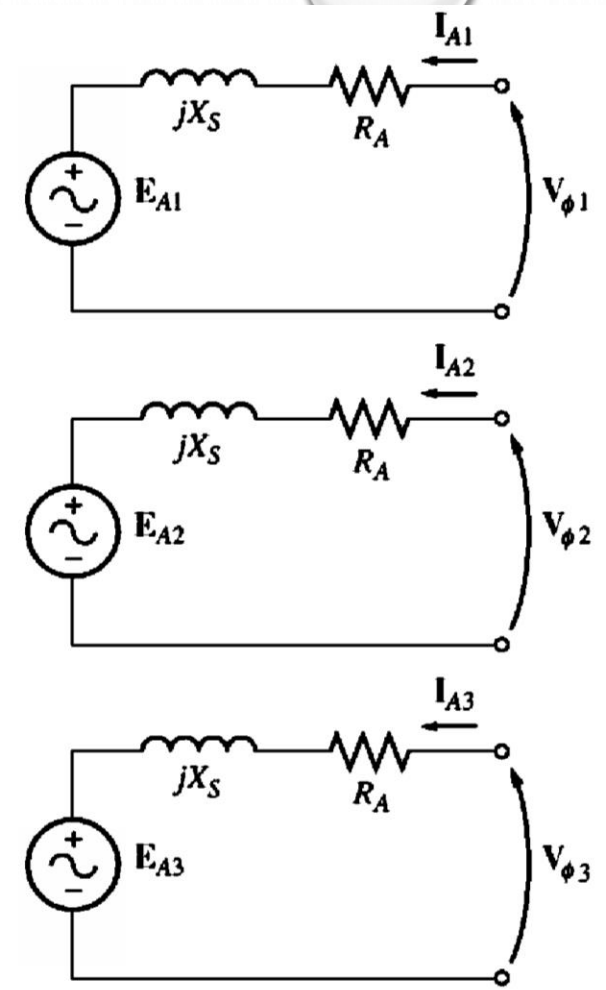
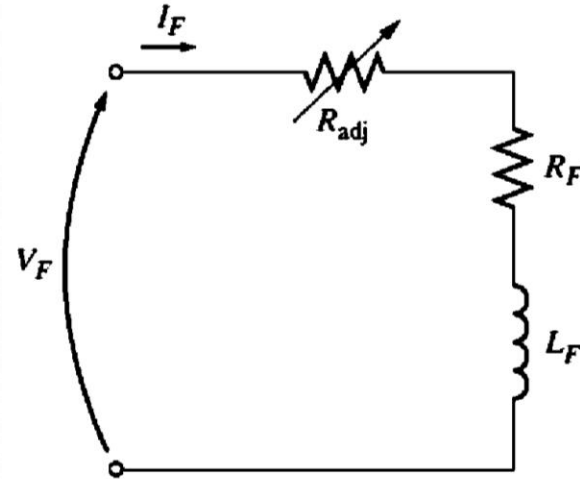
where  $\delta$  is the angle between  $\mathbf{B}_R$  and  $\mathbf{B}_{\text{net}}$ .



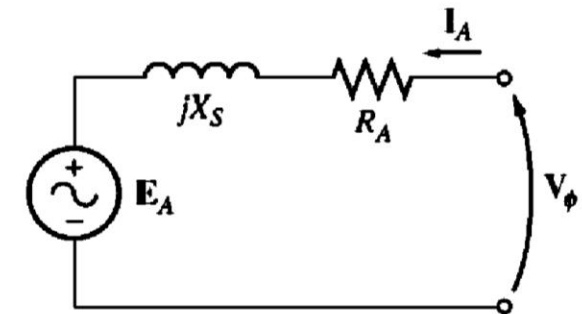
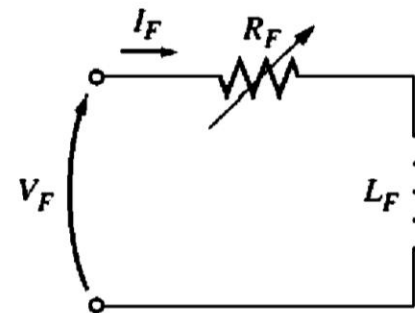
# EQUIVALENT CIRCUIT OF SM (6.1)

$$V_{\phi} = E_A + jX_S I_A + R_A I_A$$

$$E_A = V_{\phi} - jX_S I_A - R_A I_A$$



(a)

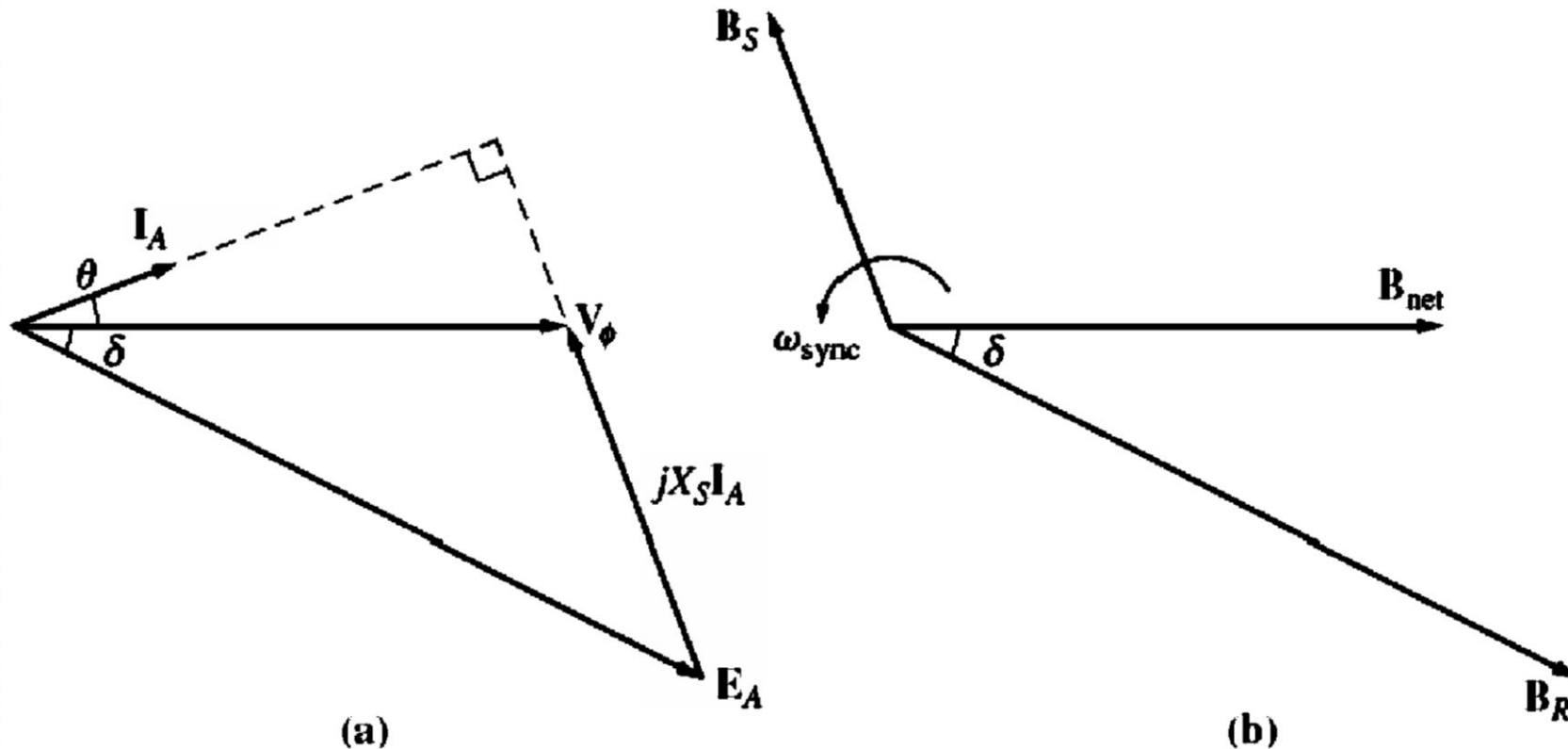


(b)



# PHASOR DIAGRAM OF SM (6.1)

$B_R$  Corresponds to produce  $E_A$   
 $B_{net}$  Corresponds to produce  $V_\phi$   
 $B_S$  Corresponds to produce  $E_{stat} (-jI_A X_S)$



**FIGURE 6-4**

(a) Phasor diagram of a synchronous motor. (b) The corresponding magnetic field diagram.

# COMPARISON OF PHASORS BETWEEN SG AND SM (6.1)

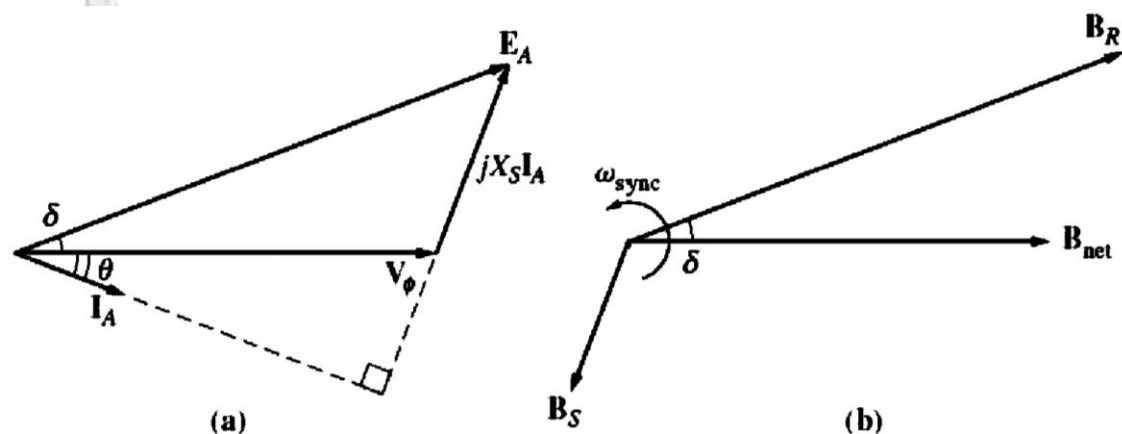


FIGURE 6-3

(a) Phasor diagram of a synchronous generator operating at a lagging power factor. (b) The corresponding magnetic field diagram.

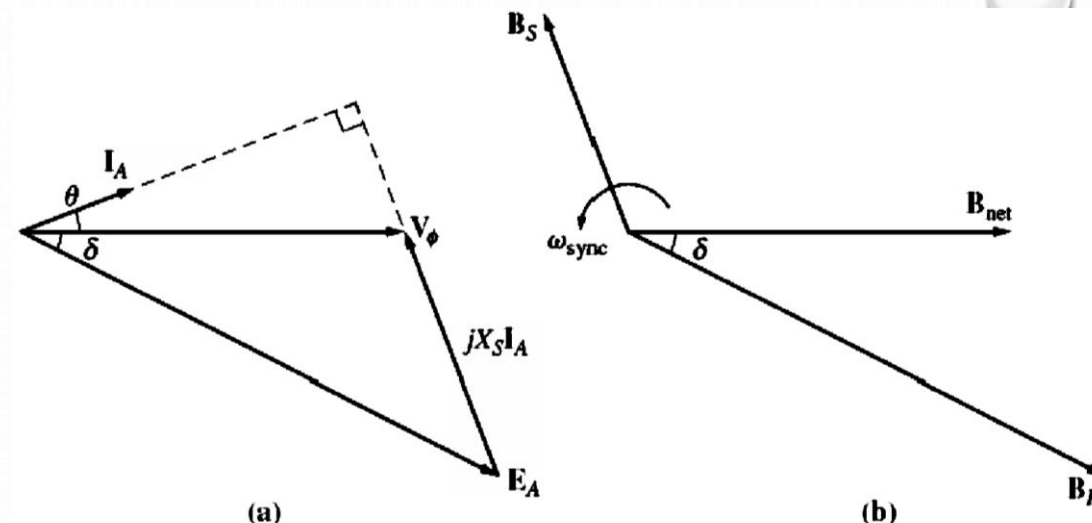


FIGURE 6-4

(a) Phasor diagram of a synchronous motor. (b) The corresponding magnetic field diagram.

$$\tau_{ind} = k \mathbf{B}_R \times \mathbf{B}_{net}$$

$$\tau_{ind} = k B_R B_{net} \sin \delta$$

The phasor diagram corresponding to generator operation is shown in Figure 6-3a, and the phasor diagram corresponding to motor operation is shown in Figure 6-4a. The reason that the quantity  $jX_S I_A$  points from  $V_\phi$  to  $E_A$  in the generator and from  $E_A$  to  $V_\phi$  in the motor is that the reference direction of  $I_A$  was reversed in the definition of the motor equivalent circuit. The basic difference between motor and generator operation in synchronous machines can be seen either in the magnetic field diagram or in the phasor diagram. *In a generator*,  $E_A$  lies ahead of  $V_\phi$ , and  $B_R$  lies ahead of  $B_{net}$ . *In a motor*,  $E_A$  lies behind  $V_\phi$ , and  $B_R$  lies behind  $B_{net}$ . In a motor the induced torque is in the direction of motion, and in a generator the induced torque is a countertorque opposing the direction of motion.

# STEADY STATE SM OPERATION (6.2)

## SM TORQUE-SPEED CHARACTERISTICS:

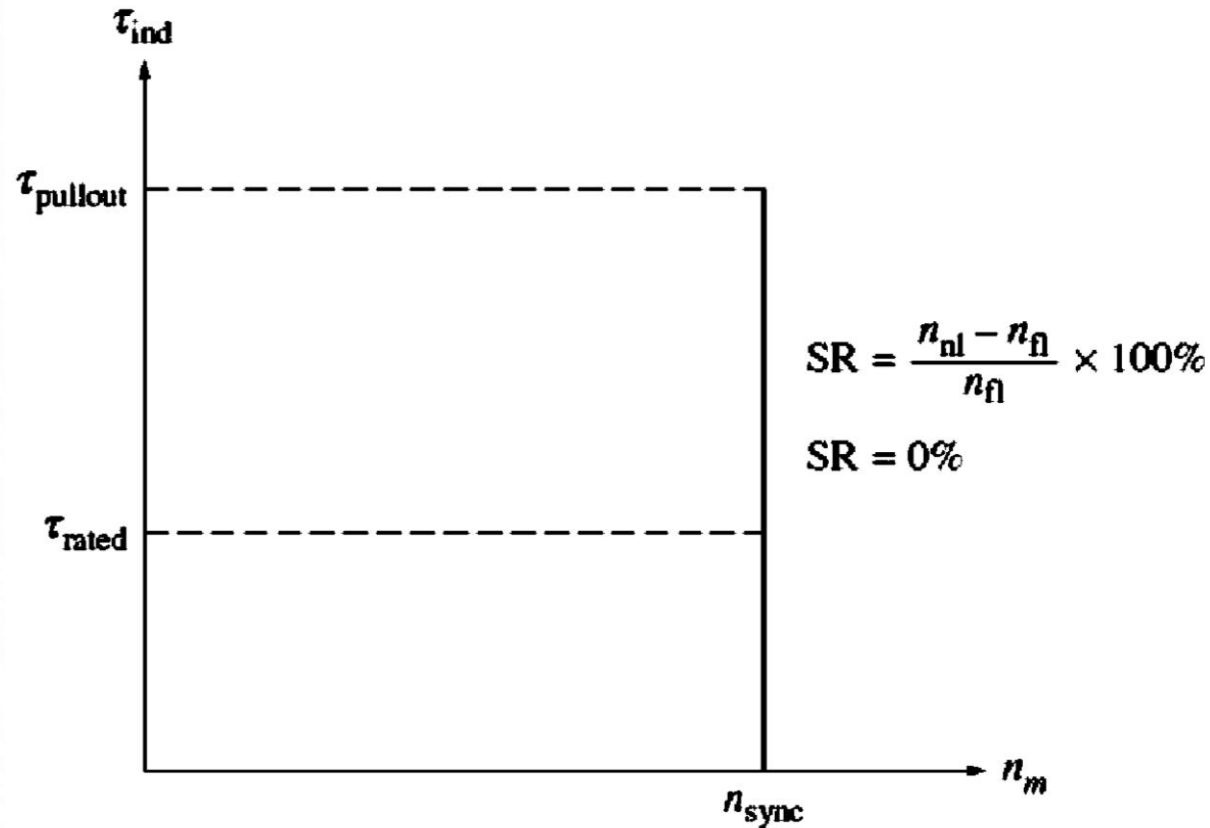


FIGURE 6-5

The torque-speed characteristic of a synchronous motor. Since the speed of the motor is constant, its speed regulation is zero.

Constant speed motor as supply voltage and frequency remain constant

$$\tau_{ind} = kB_R B_{net} \sin \delta$$

$$\tau_{max} = kB_R B_{net}$$

$$\tau_{ind} = \frac{3V_\phi E_A \sin \delta}{\omega_m X_S}$$

$$\tau_{max} = \frac{3V_\phi E_A}{\omega_m X_S}$$

The maximum or pullout torque occurs when  $\delta = 90^\circ$ . Normal full-load torques are much less than that, however. In fact, the pullout torque may typically be 3 times the full-load torque of the machine.

When the torque on the shaft of a synchronous motor exceeds the pullout torque, the rotor can no longer remain locked to the stator and net magnetic fields. Instead, the rotor starts to slip behind them. As the rotor slows down, the stator magnetic field “laps” it repeatedly, and the direction of the induced torque in the rotor reverses with each pass. The resulting huge torque surges, first one way and then the other way, cause the whole motor to vibrate severely. The loss of synchronization after the pullout torque is exceeded is known as *slipping poles*.

*These equations indicate that the larger the field current (and hence  $E_A$ ), the greater the maximum torque of the motor. There is therefore a stability advantage in operating the motor with a large field current or a large  $E_A$ .*

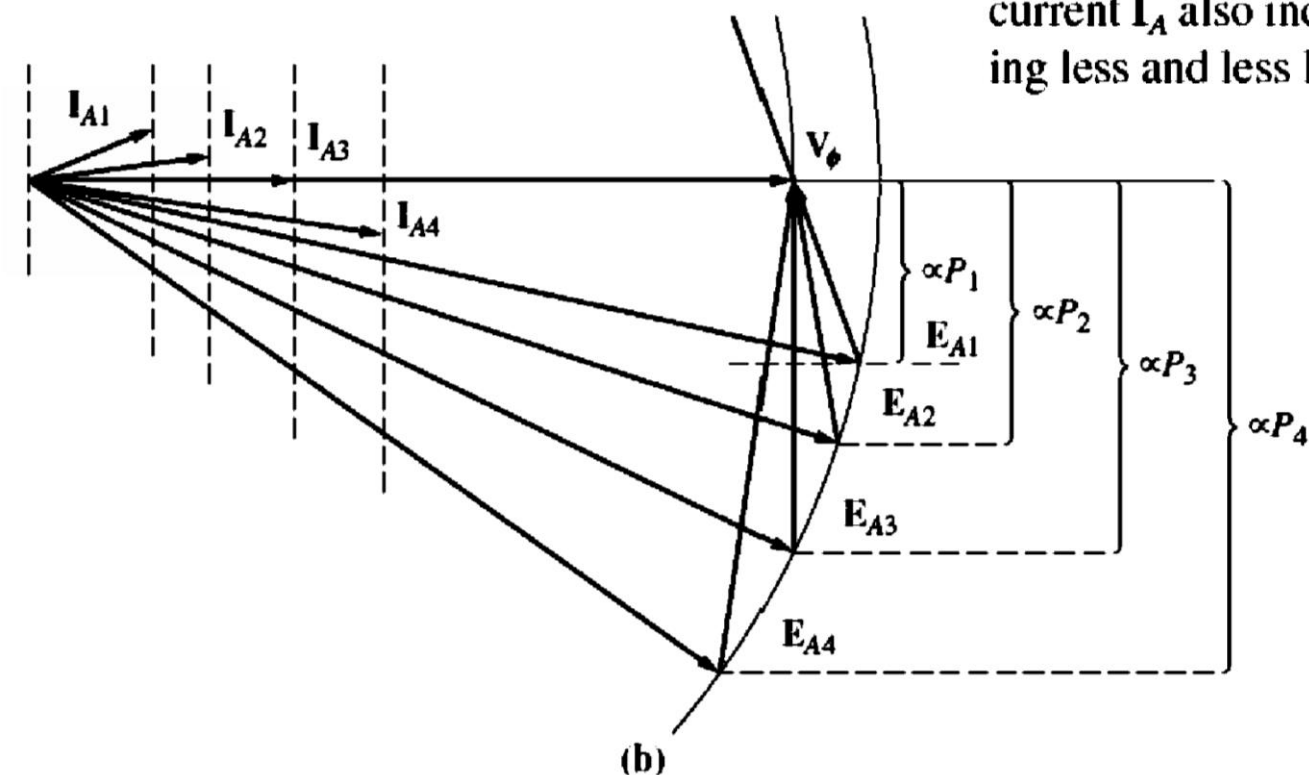
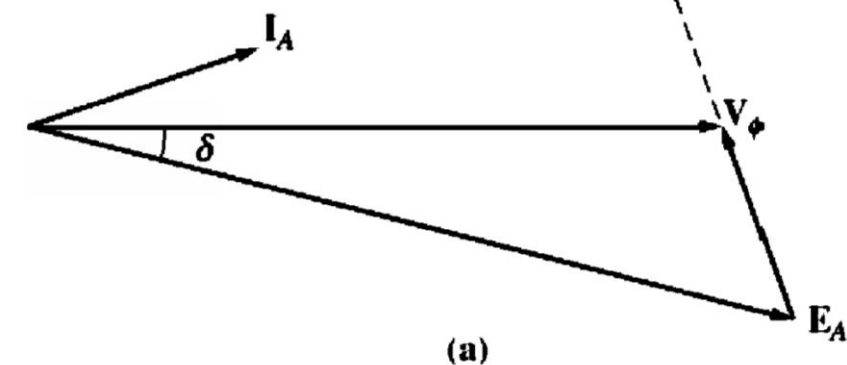


# THE EFFECT OF LOAD CHANGES ON A SM:

(Input voltage, frequency and field current constant. Only load variable)

The internal generated voltage

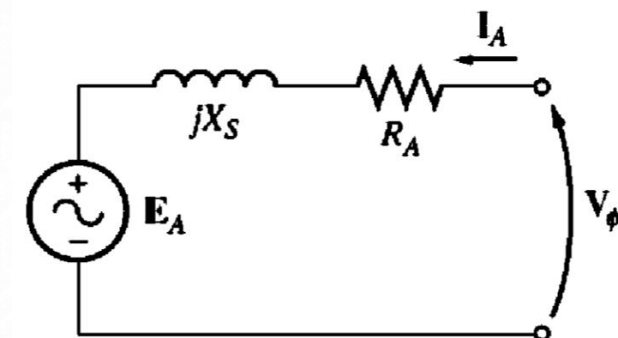
$E_A$  is equal to  $K\phi\omega$  and so depends on *only* the field current in the machine and the speed of the machine. The speed is constrained to be constant by the input power supply, and since no one has touched the field circuit, the field current is constant as well. Therefore,  $|E_A|$  *must be constant as the load changes*. The distances proportional to power ( $E_A \sin \delta$  and  $I_A \cos \theta$ ) will increase, but the magnitude of  $E_A$  must remain constant. As the load increases,  $E_A$  swings down in the manner shown in Figure 6–6b. As  $E_A$  swings down further and further, the quantity  $jX_S I_A$  has to increase to reach from the tip of  $E_A$  to  $V_\phi$ , and therefore the armature current  $I_A$  also increases. Notice that the power-factor angle  $\theta$  changes too, becoming less and less leading and then more and more lagging.



$$E_A = K\phi\omega$$

$$P = \frac{3V_\phi E_A \sin \delta}{X_S}$$

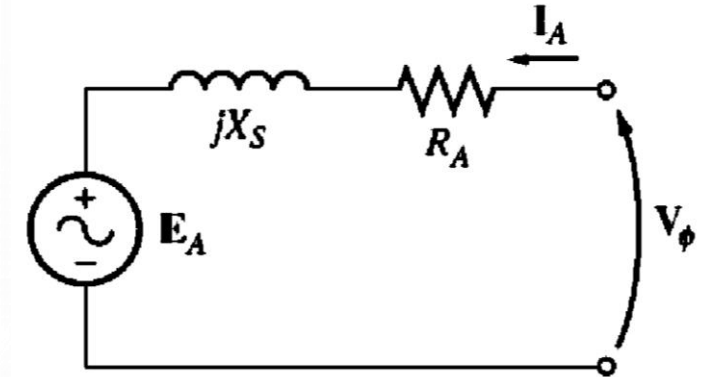
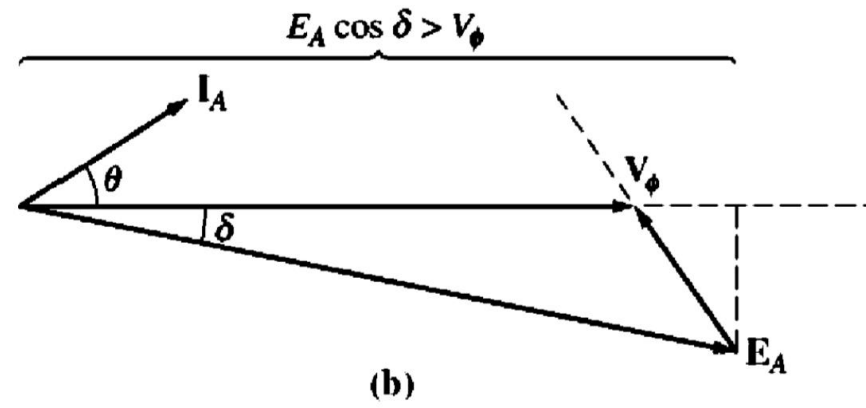
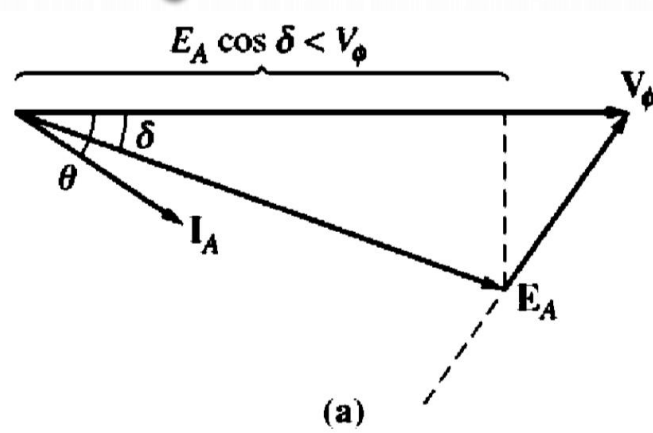
$$P = 3V_\phi I_A \cos \theta$$



# THE EFFECT OF FIELD CURRENT CHANGES ON A SM:

Ref [1]: Electric Machinery Fundamentals (4th Edition) - Stephen J. Chapman

(Input voltage, frequency and load constant. Only field current variable)

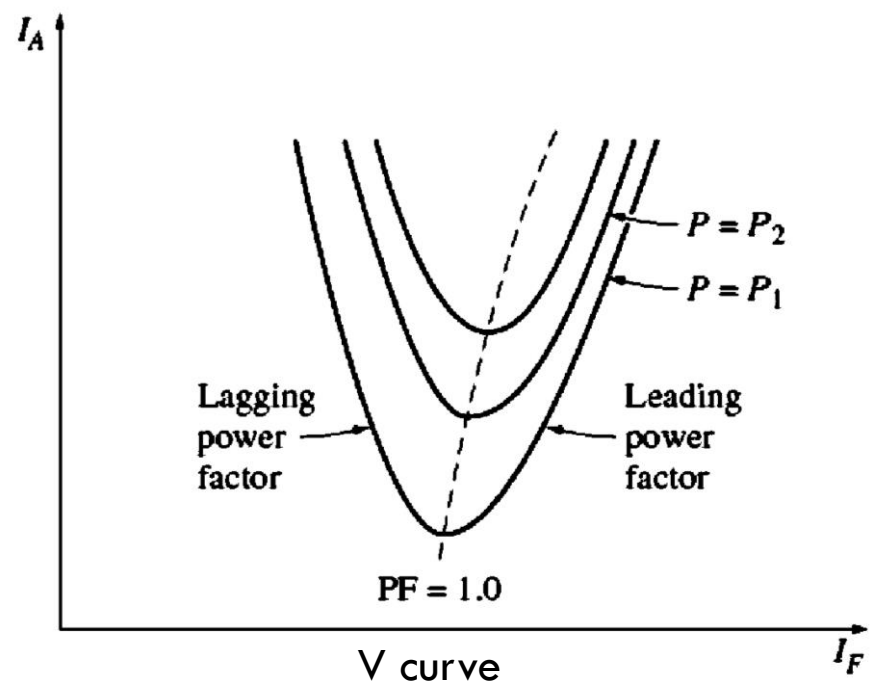


Note that *an increase in field current increases the magnitude of  $E_A$  but does not affect the real power supplied by the motor.* The power supplied by the motor changes only when the shaft load torque changes. Since a change in  $I_F$  does not affect the shaft speed  $n_m$ , and since the load attached to the shaft is unchanged, the real power supplied is unchanged. Of course,  $V_T$  is also constant, since it is kept constant by the power source supplying the motor. The distances proportional to power on the phasor diagram ( $E_A \sin \delta$  and  $I_A \cos \theta$ ) must therefore be constant. When the field current is increased,  $E_A$  must increase, but it can only do so by sliding out along the line of constant power.

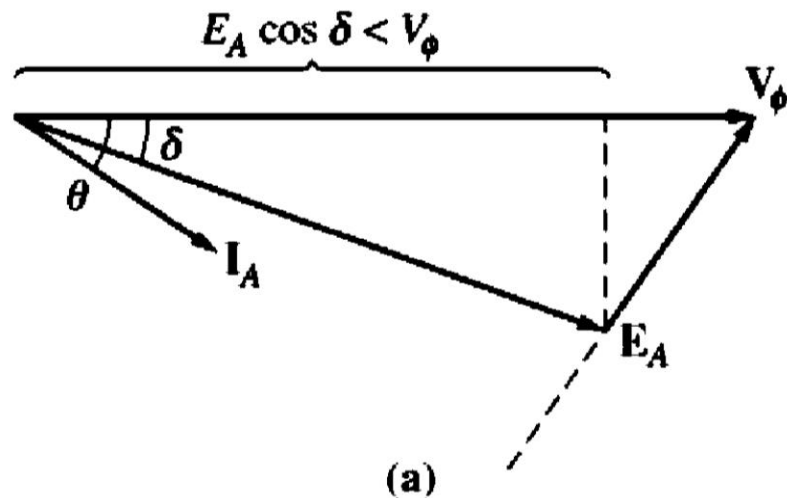
$$E_A = K\phi\omega$$

$$P = \frac{3V_\phi E_A \sin \delta}{X_S}$$

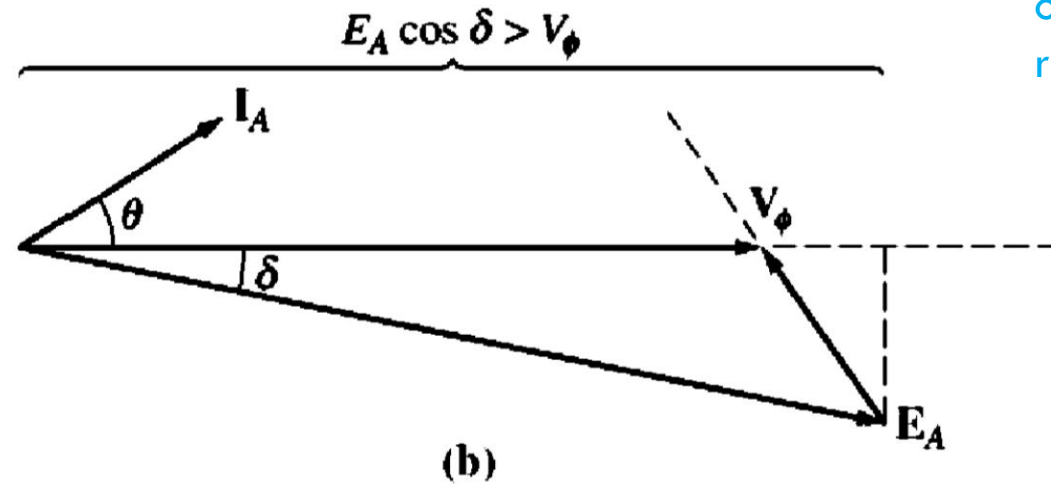
$$P = 3V_\phi I_A \cos \theta$$



Notice that as the value of  $E_A$  increases, the magnitude of the armature current  $I_A$  first decreases and then increases again. At low  $E_A$ , the armature current is lagging, and the motor is an inductive load. It is acting like an inductor-resistor combination, consuming reactive power  $Q$ . As the field current is increased, the armature current eventually lines up with  $V_\phi$ , and the motor looks purely resistive. As the field current is increased further, the armature current becomes leading, and the motor becomes a capacitive load. It is now acting like a capacitor-resistor combination, consuming negative reactive power  $-Q$  or, alternatively, supplying reactive power  $Q$  to the system.



(a) Under excited SM



(b) Over excited SM

Assignment: Draw the vector diagram of a SM when running at unity p.f.



# MATH CLASS 1

**Example 6–1.** A 208-V, 45-kVA, 0.8-PF-leading,  $\Delta$ -connected, 60-Hz synchronous machine has a synchronous reactance of  $2.5 \Omega$  and a negligible armature resistance. Its friction and windage losses are 1.5 kW, and its core losses are 1.0 kW. Initially, the shaft is supplying a 15-hp load, and the motor's power factor is 0.80 leading.

- (a) Sketch the phasor diagram of this motor, and find the values of  $I_A$ ,  $I_L$ , and  $E_A$ .
- (b) Assume that the shaft load is now increased to 30 hp. Sketch the behavior of the phasor diagram in response to this change.
- (c) Find  $I_A$ ,  $I_L$ , and  $E_A$  after the load change. What is the new motor power factor?

Solution:

- (a) Initially, the motor's output power is 15 hp. This corresponds to an output of

$$P_{\text{out}} = (15 \text{ hp})(0.746 \text{ kW/hp}) = 11.19 \text{ kW}$$

Therefore, the electric power supplied to the machine is

$$\begin{aligned} P_{\text{in}} &= P_{\text{out}} + P_{\text{mech loss}} + P_{\text{core loss}} + P_{\text{elec loss}} \\ &= 11.19 \text{ kW} + 1.5 \text{ kW} + 1.0 \text{ kW} + 0 \text{ kW} = 13.69 \text{ kW} \end{aligned}$$

Since the motor's power factor is 0.80 leading, the resulting line current flow is

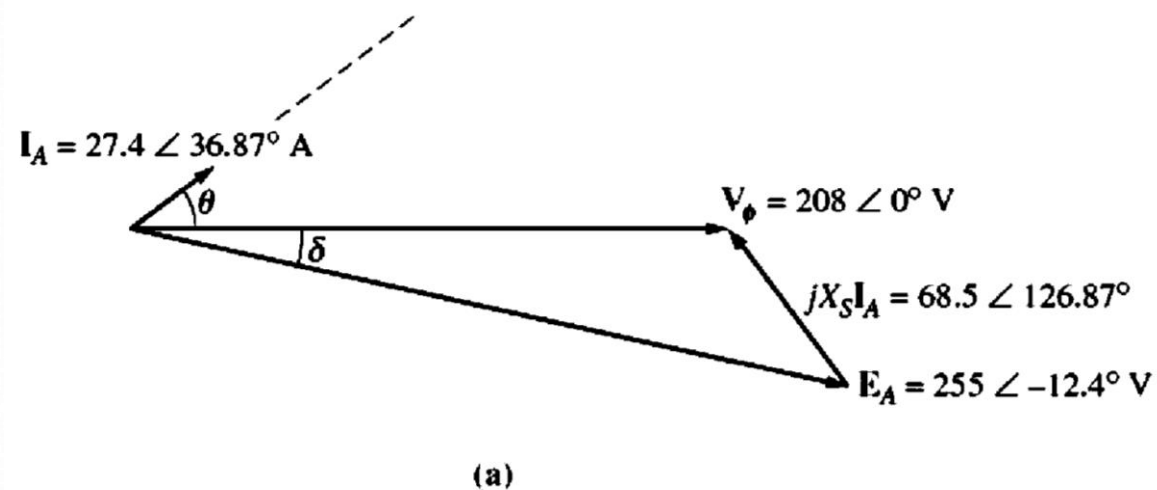
$$\begin{aligned} I_L &= \frac{P_{\text{in}}}{\sqrt{3} V_T \cos \theta} \\ &= \frac{13.69 \text{ kW}}{\sqrt{3}(208 \text{ V})(0.80)} = 47.5 \text{ A} \end{aligned}$$

and the armature current is  $I_L/\sqrt{3}$ , with 0.8 leading power factor, which gives the result

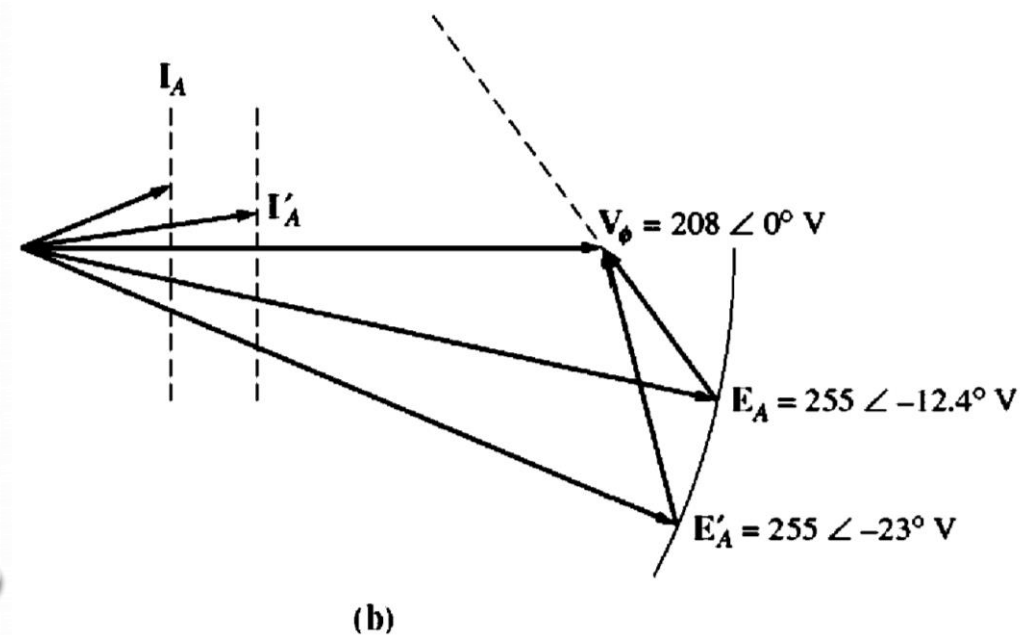
$$I_A = 27.4 \angle 36.87^\circ \text{ A}$$

To find  $E_A$ , apply Kirchhoff's voltage law [Equation (6-2)]:

$$\begin{aligned} E_A &= V_\phi - jX_S I_A \\ &= 208 \angle 0^\circ \text{ V} - (j2.5 \, \Omega)(27.4 \angle 36.87^\circ \text{ A}) \\ &= 208 \angle 0^\circ \text{ V} - 68.5 \angle 126.87^\circ \text{ V} \\ &= 249.1 - j54.8 \text{ V} = 255 \angle -12.4^\circ \text{ V} \end{aligned}$$



(b) As the power on the shaft is increased to 30 hp, the shaft slows momentarily, and the internal generated voltage  $E_A$  swings out to a larger angle  $\delta$  while maintaining a constant magnitude. The resulting phasor diagram is shown in Figure 6-7b.



(c) After the load changes, the electric input power of the machine becomes

$$\begin{aligned}
 P_{\text{in}} &= P_{\text{out}} + P_{\text{mech loss}} + P_{\text{core loss}} + P_{\text{elec loss}} \\
 &= (30 \text{ hp})(0.746 \text{ kW/hp}) + 1.5 \text{ kW} + 1.0 \text{ kW} + 0 \text{ kW} \\
 &= 24.88 \text{ kW}
 \end{aligned}$$

From the equation for power in terms of torque angle [Equation (5-20)], it is possible to find the magnitude of the angle  $\delta$  (remember that the magnitude of  $E_A$  is constant):

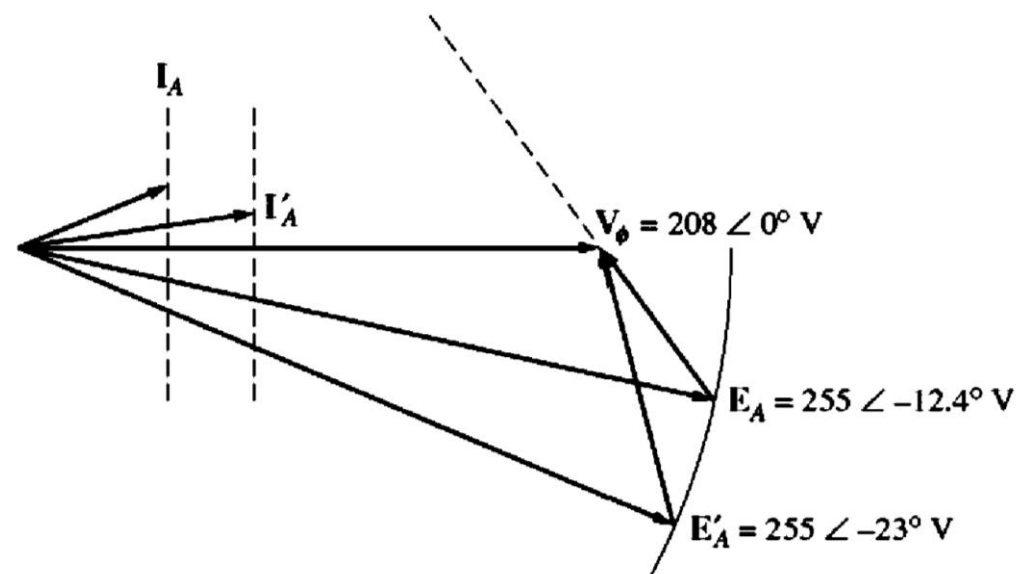
$$P = \frac{3V_\phi E_A \sin \delta}{X_S} \quad (5-20)$$

so

$$\begin{aligned}
 \delta &= \sin^{-1} \frac{X_S P}{3V_\phi E_A} \\
 &= \sin^{-1} \frac{(2.5 \Omega)(24.88 \text{ kW})}{3(208 \text{ V})(255 \text{ V})} \\
 &= \sin^{-1} 0.391 = 23^\circ
 \end{aligned}$$

The internal generated voltage thus becomes  $E_A = 255 \angle -23^\circ \text{ V}$ . Therefore,  $I_A$  will be given by

$$\begin{aligned}
 I_A &= \frac{V_\phi - E_A}{jX_S} \\
 &= \frac{208 \angle 0^\circ \text{ V} - 255 \angle -23^\circ \text{ V}}{j2.5 \Omega}
 \end{aligned}$$



$$= \frac{103.1 \angle 105^\circ \text{ V}}{j2.5 \Omega} = 41.2 \angle 15^\circ \text{ A}$$

and  $I_L$  will become

$$I_L = \sqrt{3} I_A = 71.4 \text{ A}$$

The final power factor will be  $\cos(-15^\circ)$  or 0.966 leading.



**Example 6–2.** The 208-V, 45-kVA, 0.8-PF-leading,  $\Delta$ -connected, 60-Hz synchronous motor of the previous example is supplying a 15-hp load with an initial power factor of 0.85 PF lagging. The field current  $I_F$  at these conditions is 4.0 A.

- (a) Sketch the initial phasor diagram of this motor, and find the values  $I_A$  and  $E_A$ .
- (b) If the motor's flux is increased by 25 percent, sketch the new phasor diagram of the motor. What are  $E_A$ ,  $I_A$ , and the power factor of the motor now?
- (c) Assume that the flux in the motor varies linearly with the field current  $I_F$ . Make a plot of  $I_A$  versus  $I_F$  for the synchronous motor with a 15-hp load. (FYI)

Solution:

- (a) Initially, the motor's output power is 15 hp. This corresponds to an output of

$$P_{\text{out}} = (15 \text{ hp})(0.746 \text{ kW/hp}) = 11.19 \text{ kW}$$

Therefore, the electric power supplied to the machine is

$$\begin{aligned} P_{\text{in}} &= P_{\text{out}} + P_{\text{mech loss}} + P_{\text{core loss}} + P_{\text{elec loss}} \\ &= 11.19 \text{ kW} + 1.5 \text{ kW} + 1.0 \text{ kW} + 0 \text{ kW} = 13.69 \text{ kW} \end{aligned}$$

Since the motor's power factor is 0.85 lagging, the resulting armature current flow is

$$\begin{aligned} I_A &= \frac{P_{\text{in}}}{3V_{\phi} \cos \theta} \\ &= \frac{13.69 \text{ kW}}{3(208 \text{ V})(0.85)} = 25.8 \text{ A} \end{aligned}$$

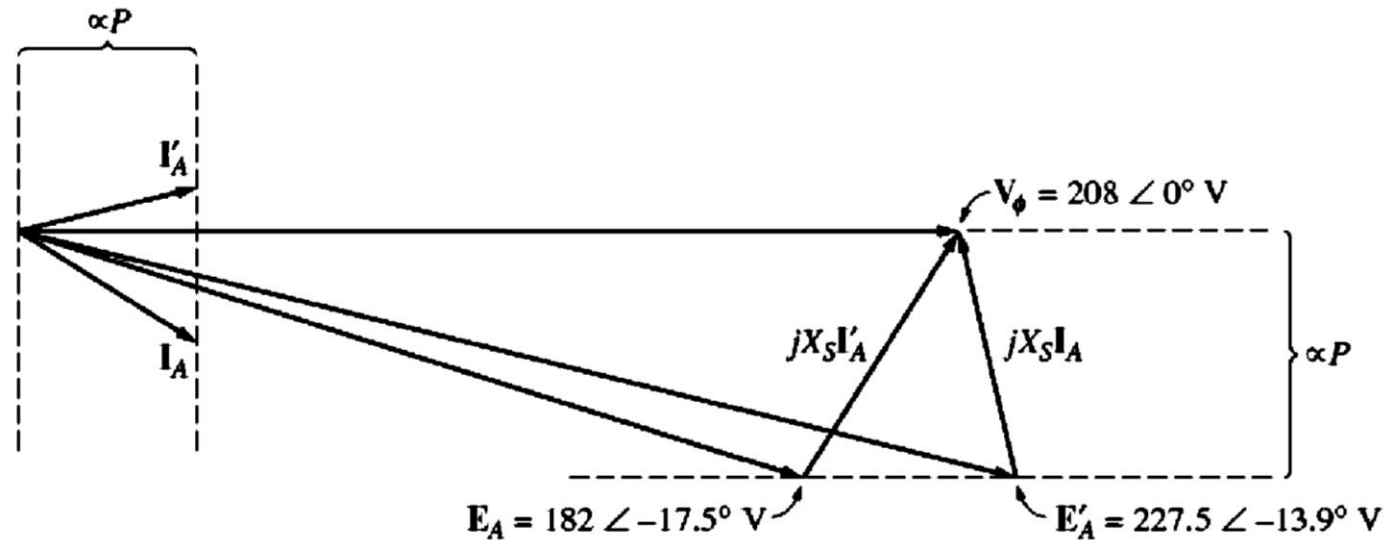
The angle  $\theta$  is  $\cos^{-1} 0.85 = 31.8^\circ$ , so the phasor current  $I_A$  is equal to

$$I_A = 25.8 \angle -31.8^\circ \text{ A}$$

To find  $E_A$ , apply Kirchhoff's voltage law [Equation (6-2)]:

$$\begin{aligned} E_A &= V_\phi - jX_S I_A \\ &= 208 \angle 0^\circ \text{ V} - (j2.5 \Omega)(25.8 \angle -31.8^\circ \text{ A}) \\ &= 208 \angle 0^\circ \text{ V} - 64.5 \angle 58.2^\circ \text{ V} \\ &= 182 \angle -17.5^\circ \text{ V} \end{aligned}$$

The resulting phasor diagram is shown in Figure 6-11, together with for part *b*.



(*b*) If the flux  $\phi$  is increased by 25 percent, then  $E_A = K\phi\omega$  will increase by 25 percent too:

$$E_{A2} = 1.25 E_{A1} = 1.25(182 \text{ V}) = 227.5 \text{ V}$$

However, the power supplied to the load must remain constant. Since the distance  $E_A \sin \delta$  is proportional to the power, that distance on the phasor diagram must be constant from the original flux level to the new flux level. Therefore,

$$\begin{aligned} E_{A1} \sin \delta_1 &= E_{A2} \sin \delta_2 \\ \delta_2 &= \sin^{-1} \left( \frac{E_{A1}}{E_{A2}} \sin \delta_1 \right) \\ &= \sin^{-1} \left[ \frac{182 \text{ V}}{227.5 \text{ V}} \sin (-17.5^\circ) \right] = -13.9^\circ \end{aligned}$$

The armature current can now be found from Kirchhoff's voltage law:

$$\begin{aligned} I_{A2} &= \frac{V_\phi - E_{A2}}{jX_S} \\ I_A &= \frac{208 \angle 0^\circ \text{ V} - 227.5 \angle -13.9^\circ \text{ V}}{j2.5 \Omega} \\ &= \frac{56.2 \angle 103.2^\circ \text{ V}}{j2.5 \Omega} = 22.5 \angle 13.2^\circ \text{ A} \end{aligned}$$

Finally, the motor's power factor is now

$$\text{PF} = \cos (13.2^\circ) = 0.974 \quad \text{leading}$$

The resulting phasor diagram is also shown in Figure 6-11.

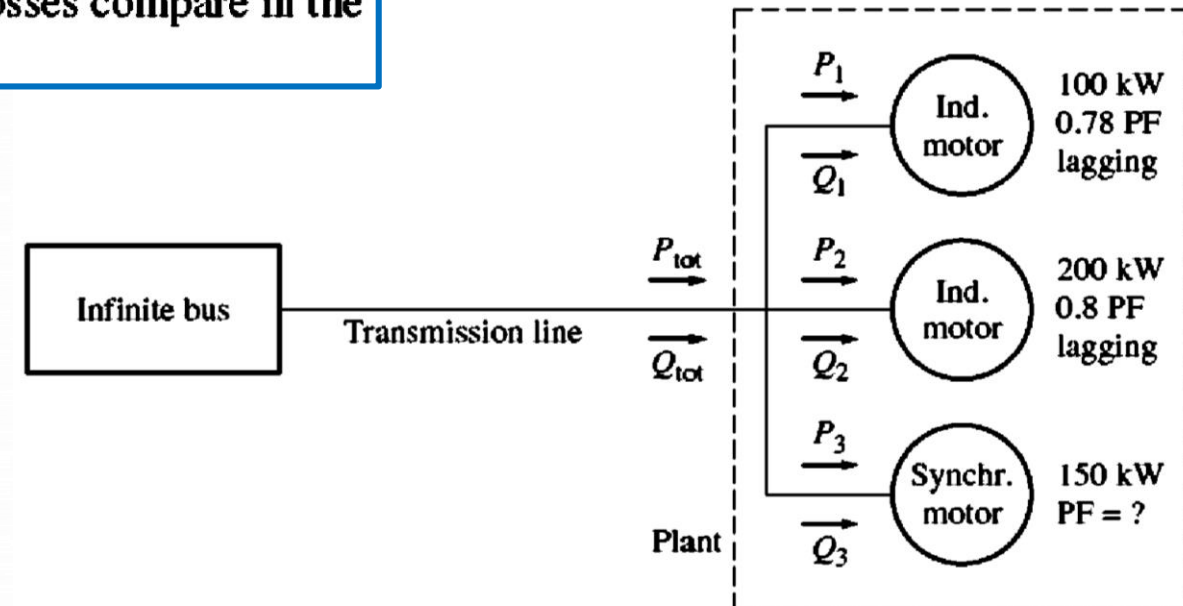
## SM and the power factor correction:

**Example 6–3.** The infinite bus in Figure 6–13 operates at 480 V. Load 1 is an induction motor consuming 100 kW at 0.78 PF lagging, and load 2 is an induction motor consuming 200 kW at 0.8 PF lagging. Load 3 is a synchronous motor whose real power consumption is 150 kW.

- (a) If the synchronous motor is adjusted to operate at 0.85 PF lagging, what is the transmission line current in this system?
- (b) If the synchronous motor is adjusted to operate at 0.85 PF leading, what is the transmission line current in this system?
- (c) Assume that the transmission line losses are given by

$$P_{LL} = 3I_L^2 R_L \quad \text{line loss}$$

where LL stands for line losses. How do the transmission losses compare in the two cases?





- (a) In the first case, the real power of load 1 is 100 kW, and the reactive power of load 1 is

$$\begin{aligned} Q_1 &= P_1 \tan \theta \\ &= (100 \text{ kW}) \tan (\cos^{-1} 0.78) = (100 \text{ kW}) \tan 38.7^\circ \\ &= 80.2 \text{ kVAR} \end{aligned}$$

The real power of load 2 is 200 kW, and the reactive power of load 2 is

$$\begin{aligned} Q_2 &= P_2 \tan \theta \\ &= (200 \text{ kW}) \tan (\cos^{-1} 0.80) = (200 \text{ kW}) \tan 36.87^\circ \\ &= 150 \text{ kVAR} \end{aligned}$$

The real power load 3 is 150 kW, and the reactive power of load 3 is

$$\begin{aligned} Q_3 &= P_3 \tan \theta \\ &= (150 \text{ kW}) \tan (\cos^{-1} 0.85) = (150 \text{ kW}) \tan 31.8^\circ \\ &= 93 \text{ kVAR} \end{aligned}$$

Thus, the total real load is

$$\begin{aligned} P_{\text{tot}} &= P_1 + P_2 + P_3 \\ &= 100 \text{ kW} + 200 \text{ kW} + 150 \text{ kW} = 450 \text{ kW} \end{aligned}$$

and the total reactive load is

$$\begin{aligned} Q_{\text{tot}} &= Q_1 + Q_2 + Q_3 \\ &= 80.2 \text{ kVAR} + 150 \text{ kVAR} + 93 \text{ kVAR} = 323.2 \text{ kVAR} \end{aligned}$$

The equivalent system power factor is thus

$$\begin{aligned} \text{PF} &= \cos \theta = \cos \left( \tan^{-1} \frac{Q}{P} \right) = \cos \left( \tan^{-1} \frac{323.2 \text{ kVAR}}{450 \text{ kW}} \right) \\ &= \cos 35.7^\circ = 0.812 \text{ lagging} \end{aligned}$$

Finally, the line current is given by

$$I_L = \frac{P_{\text{tot}}}{\sqrt{3} V_L \cos \theta} = \frac{450 \text{ kW}}{\sqrt{3}(480 \text{ V})(0.812)} = 667 \text{ A}$$

- (b) The real and reactive powers of loads 1 and 2 are unchanged, as is the real power of load 3. The reactive power of load 3 is

$$\begin{aligned} Q_3 &= P_3 \tan \theta \\ &= (150 \text{ kW}) \tan (-\cos^{-1} 0.85) = (150 \text{ kW}) \tan (-31.8^\circ) \\ &= -93 \text{ kVAR} \end{aligned}$$

Thus, the total real load is

$$\begin{aligned} P_{\text{tot}} &= P_1 + P_2 + P_3 \\ &= 100 \text{ kW} + 200 \text{ kW} + 150 \text{ kW} = 450 \text{ kW} \end{aligned}$$

and the total reactive load is

$$\begin{aligned} Q_{\text{tot}} &= Q_1 + Q_2 + Q_3 \\ &= 80.2 \text{ kVAR} + 150 \text{ kVAR} - 93 \text{ kVAR} = 137.2 \text{ kVAR} \end{aligned}$$

The equivalent system power factor is thus

$$\begin{aligned} \text{PF} &= \cos \theta = \cos \left( \tan^{-1} \frac{Q}{P} \right) = \cos \left( \tan^{-1} \frac{137.2 \text{ kVAR}}{450 \text{ kW}} \right) \\ &= \cos 16.96^\circ = 0.957 \text{ lagging} \end{aligned}$$

Finally, the line current is given by

$$I_L = \frac{P_{\text{tot}}}{\sqrt{3} V_L \cos \theta} = \frac{450 \text{ kW}}{\sqrt{3}(480 \text{ V})(0.957)} = 566 \text{ A}$$

- (c) The transmission losses in the first case are

$$P_{\text{LL}} = 3I_L^2 R_L = 3(667 \text{ A})^2 R_L = 1,344,700 R_L$$

The transmission losses in the second case are

$$P_{\text{LL}} = 3I_L^2 R_L = 3(566 \text{ A})^2 R_L = 961,070 R_L$$

Notice that in the second case the transmission power losses are 28 percent less than in the first case, while the power supplied to the loads is the same.

The use of synchronous motors or other equipment to increase the overall power factor of a power system is called *power-factor correction*.

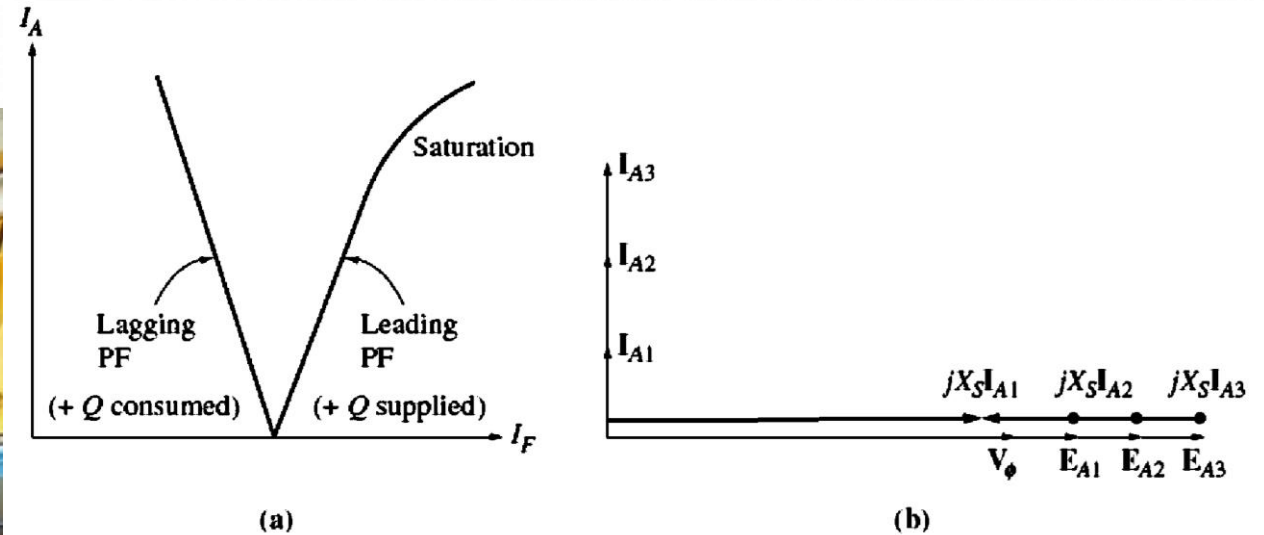
1. A leading load can supply some reactive power  $Q$  for nearby lagging loads, instead of it coming from the generator. Since the reactive power does not have to travel over the long and fairly high-resistance transmission lines, the transmission line current is reduced and the power system losses are much lower. (This was shown by the previous example.)
2. Since the transmission lines carry less current, they can be smaller for a given rated power flow. A lower equipment current rating reduces the cost of a power system significantly.
3. In addition, requiring a synchronous motor to operate with a leading power factor means that the motor must be run *overexcited*. This mode of operation increases the motor's maximum torque and reduces the chance of accidentally exceeding the pullout torque.



# SYNCHRONOUS CAPACITOR OR SYNCHRONOUS CONDENSER



Ref [4]: <https://www.gepowerconversion.com/press-releases/largest-motor-ever-produced-peterborough-out-door>



Some synchronous motors used to be sold specifically for power-factor correction. These machines had shafts that did not even come through the frame of the motor—no load could be connected to them even if one wanted to do so. Such special-purpose synchronous motors were often called *synchronous condensers* or *synchronous capacitors*. (*Condenser* is an old name for capacitor.)

The V curve for a synchronous capacitor is shown in Figure 6–15a. Since the real power supplied to the machine is zero (except for losses), at unity power factor the current  $I_A = 0$ . As the field current is increased above that point, the line current (and the reactive power supplied by the motor) increases in a nearly linear fashion until saturation is reached. Figure 6–15b shows the effect of increasing the field current on the motor's phasor diagram.

Today, conventional static capacitors are more economical to buy and use than synchronous capacitors. However, some synchronous capacitors may still be in use in older industrial plants.



# STARTING OF SM (6.3)

Why SM not self starting:

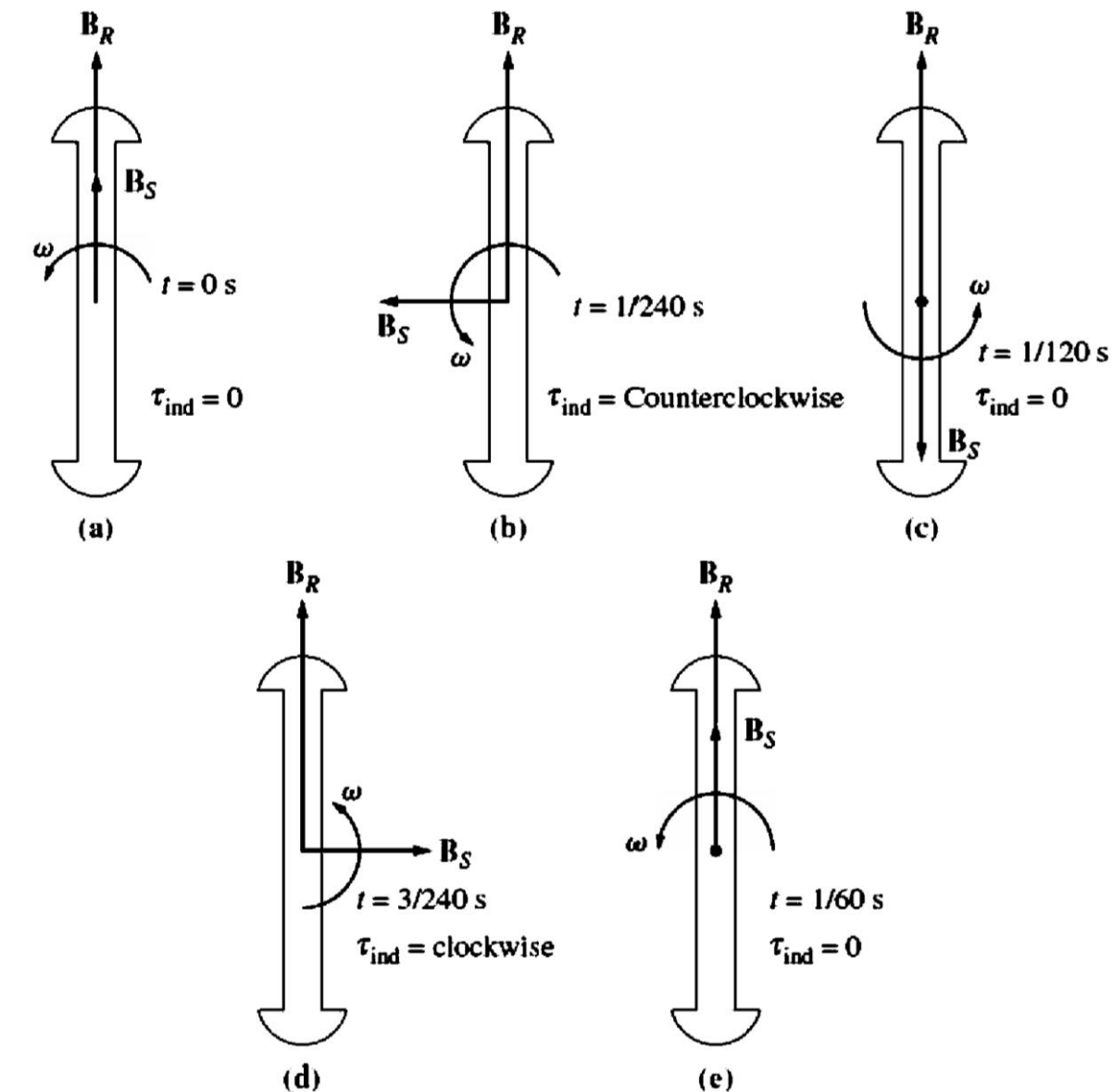


Figure 6–16a shows the machine at time  $t = 0$  s, when  $B_R$  and  $B_S$  are exactly lined up. By the induced-torque equation

$$\tau_{ind} = k B_R \times B_S \quad (4-58)$$

the induced torque on the shaft of the rotor is zero. Figure 6–16b shows the situation at time  $t = 1/240$  s. In such a short time, the rotor has barely moved, but the stator magnetic field now points to the left. By the induced-torque equation, the torque on the shaft of the rotor is now *counterclockwise*. Figure 6–16c shows the situation at time  $t = 1/120$  s. At that point  $B_R$  and  $B_S$  point in opposite directions, and  $\tau_{ind}$  again equals zero. At  $t = 1/60$  s, the stator magnetic field now points to the right, and the resulting torque is *clockwise*.

Finally, at  $t = 1/60$  s, the stator magnetic field is again lined up with the rotor magnetic field, and  $\tau_{ind} = 0$ . During one electrical cycle, the torque was first counterclockwise and then clockwise, and the average torque over the complete cycle was zero. What happens to the motor is that it vibrates heavily with each electrical cycle and finally overheats.

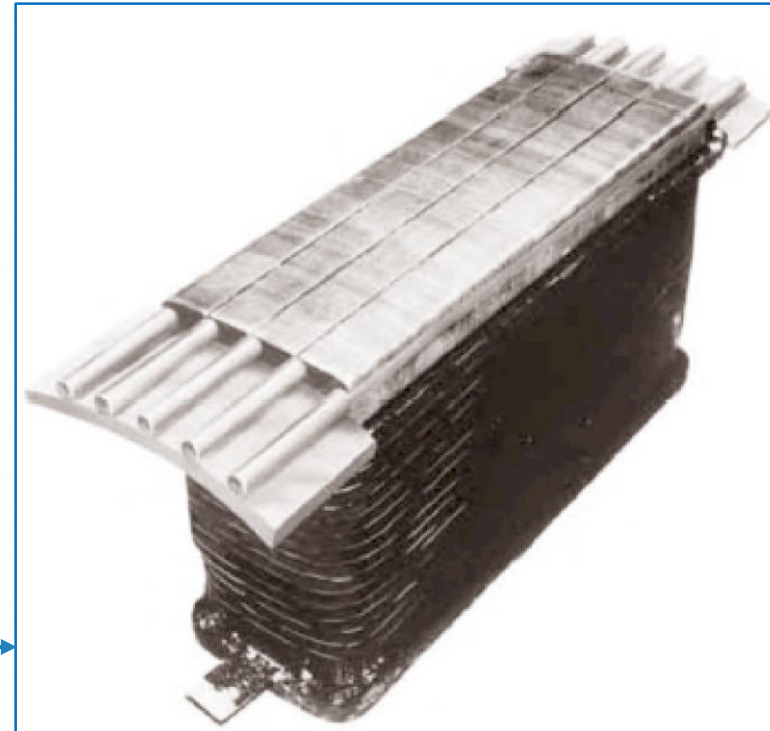
# Methods of making a SM self starting:

1. *Reduce the speed of the stator magnetic field* to a low enough value that the rotor can accelerate and lock in with it during one half-cycle of the magnetic field's rotation. This can be done by reducing the frequency of the applied electric power.
2. *Use an external prime mover* to accelerate the synchronous motor up to synchronous speed, go through the paralleling procedure, and bring the machine on the line as a generator. Then, turning off or disconnecting the prime mover will make the synchronous machine a motor.

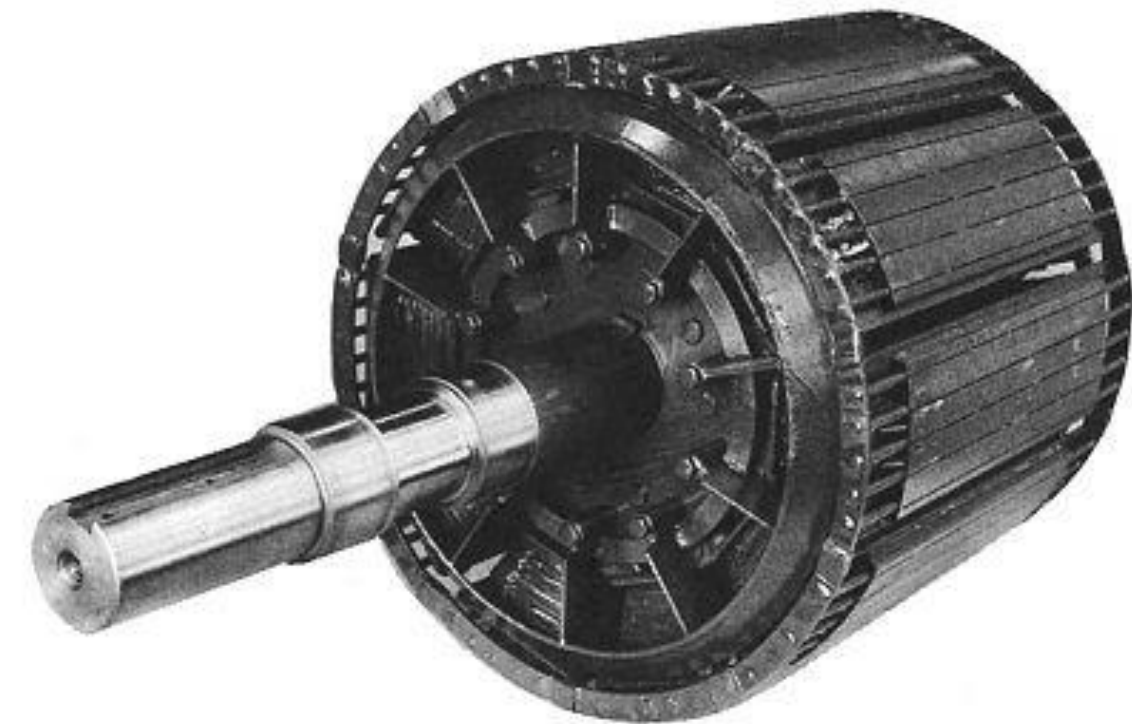
- Variable frequency drive (VFD) (rectifier-inverter or cycloconverter) is needed
- For low frequency start, input voltage to SM must be lowered in order to keep the stator current at safe level

The second approach to starting a synchronous motor is to attach an external starting motor to it and bring the synchronous machine up to full speed with the external motor. Then the synchronous machine can be paralleled with its power system as a generator, and the starting motor can be detached from the shaft of the machine. Once the starting motor is turned off, the shaft of the machine slows down, the rotor magnetic field  $\mathbf{B}_R$  falls behind  $\mathbf{B}_{net}$ , and the synchronous machine starts to act as a motor. Once paralleling is completed, the synchronous motor can be loaded down in an ordinary fashion.

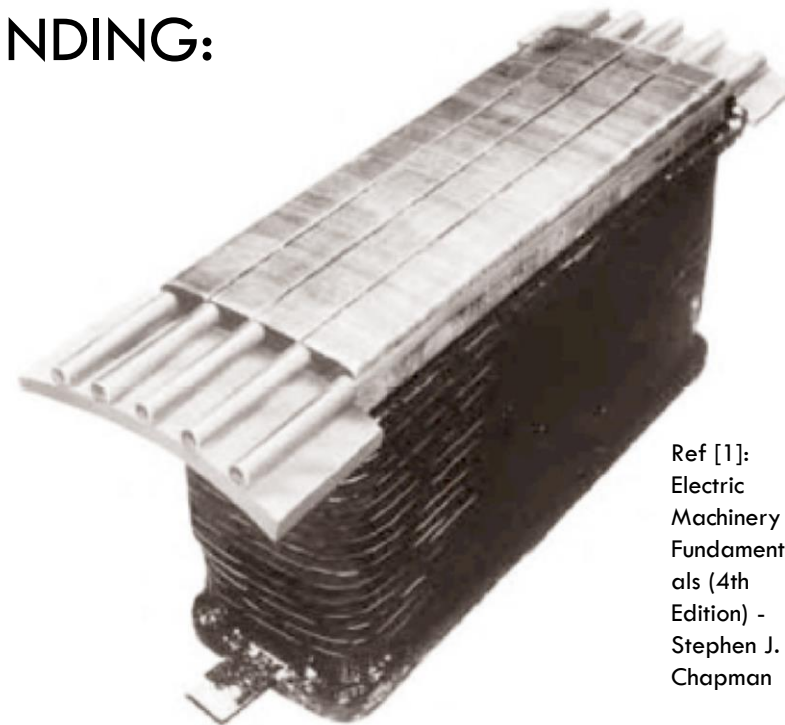
3. *Use damper windings or amortisseur windings.* The function of damper windings and their use in motor starting will be explained below.



# SM STARTING BY USING AMORTISSEUR (DAMPER) WINDING:



Ref [5]: [https://www.industrial-electronics.com/Motor\\_Control\\_8-39.html](https://www.industrial-electronics.com/Motor_Control_8-39.html)

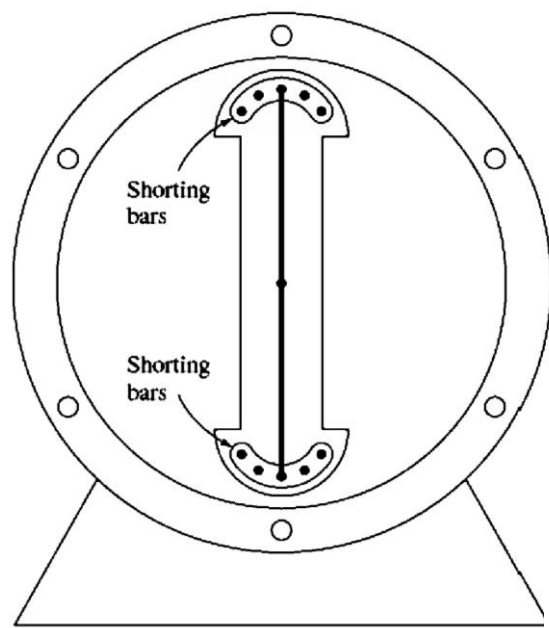


Ref [1]:  
Electric  
Machinery  
Fundamentals (4th  
Edition) -  
Stephen J.  
Chapman



Ref [6]: <https://electrical-engineering-portal.com/synchronous-motor-construction>





$$e_{\text{ind}} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l}$$

$\mathbf{v}$  = velocity of the bar *relative to the magnetic field*

$\mathbf{B}$  = magnetic flux density vector

$\mathbf{l}$  = length of conductor in the magnetic field

$$\text{At } t = 0, \quad \tau_{\text{ind}} = k\mathbf{B}_w \times \mathbf{B}_s$$

the resulting torque on the bars (and the rotor) is *counterclockwise*.

Figure 6–19b shows the situation at  $t = 1/240$  s. Here, the stator magnetic field has rotated  $90^\circ$  while the rotor has barely moved (it simply cannot speed up in so short a time). At this point, the voltage induced in the amortisseur windings is zero, because  $\mathbf{v}$  is parallel to  $\mathbf{B}$ . With no induced voltage, there is no current in the windings, and the induced torque is zero.

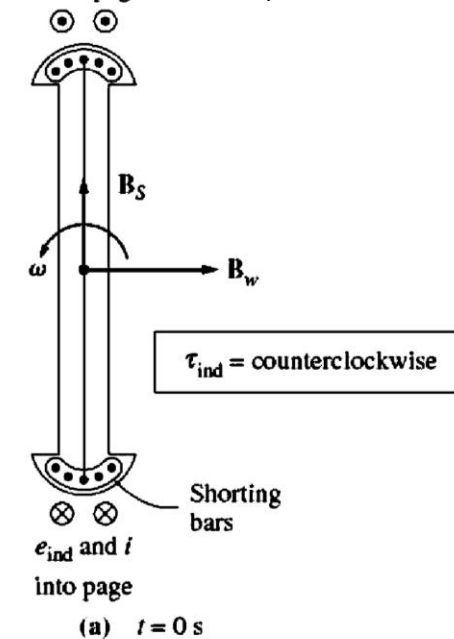
Figure 6–19c shows the situation at  $t = 1/120$  s. Now the stator magnetic field has rotated  $90^\circ$ , and the rotor still has not moved yet. The induced voltage [given by Equation (1–45)] in the amortisseur windings is out of the page in the bottom bars and into the page in the top bars. The resulting current flow is out of the page in the bottom bars and into the page in the top bars, causing a magnetic field  $\mathbf{B}_w$  to point to the left. The resulting induced torque, given by

$$\tau_{\text{ind}} = k\mathbf{B}_w \times \mathbf{B}_s$$

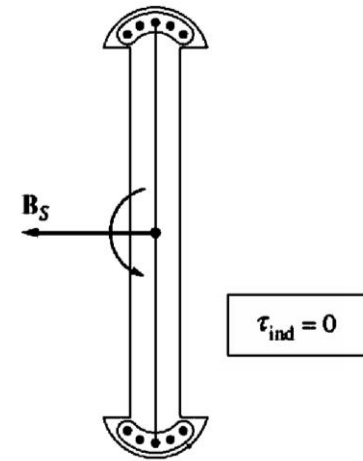
is counterclockwise.

Finally, Figure 6–19d shows the situation at time  $t = 3/240$  s. Here, as at  $t = 1/240$  s, the induced torque is zero.

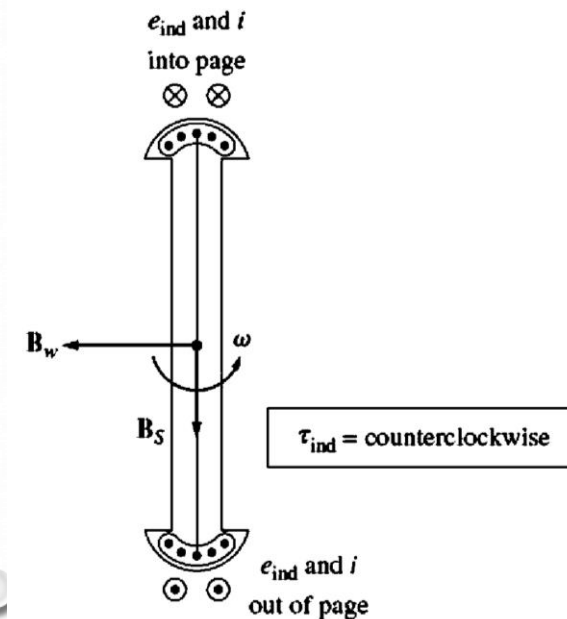
$e_{\text{ind}}$  and  $i$   
out of page



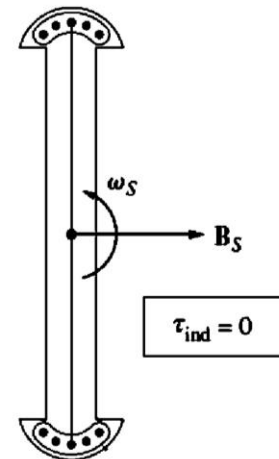
(a)  $t = 0$  s



(b)  $t = 1/240$  s



(c)  $t = 1/120$  s



(d)  $t = 3/240$  s

In summary the process of starting SM by amortisseur winding:

Effect of amortisseur winding in Motor stability:

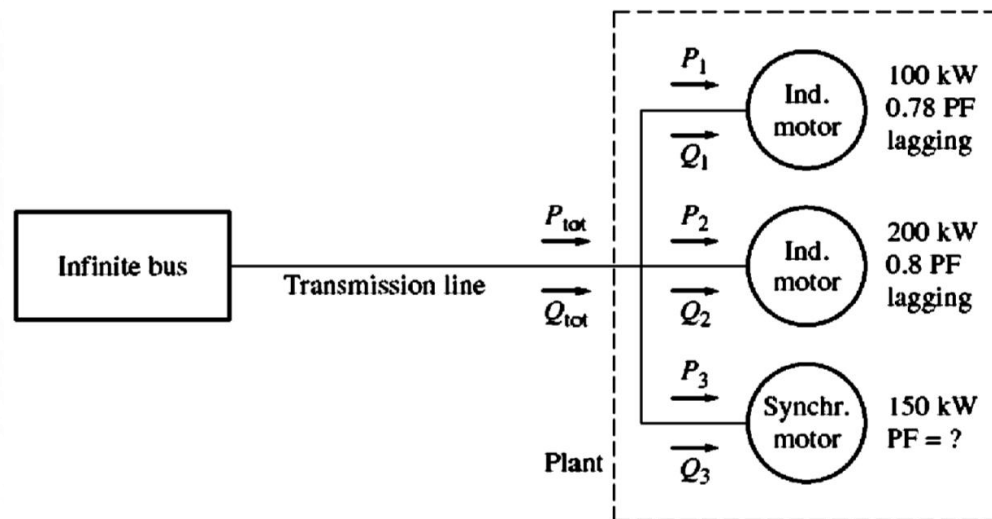
1. Disconnect the field windings from their dc power source and short them out.
2. Apply a three-phase voltage to the stator of the motor, and let the rotor accelerate up to near-synchronous speed. The motor should have no load on its shaft, so that its speed can approach  $n_{\text{sync}}$  as closely as possible.
3. Connect the dc field circuit to its power source. After this is done, the motor will lock into step at synchronous speed, and loads may then be added to its shaft.

a free bonus—an increase in machine stability. The stator magnetic field rotates at a constant speed  $n_{\text{sync}}$ , which varies only when the system frequency varies. If the rotor turns at  $n_{\text{sync}}$ , then the amortisseur windings have no induced voltage at all. If the rotor turns *slower* than  $n_{\text{sync}}$ , then there will be relative motion between the rotor and the stator magnetic field and a voltage will be induced in the windings. This voltage produces a current flow, and the current flow produces a magnetic field. The interaction of the two magnetic fields produces a torque that tends to speed the machine up again. On the other hand, if the rotor turns *faster* than the stator magnetic field, a torque will be produced that tries to slow the rotor down. Thus, *the torque produced by the amortisseur windings speeds up slow machines and slows down fast machines.*

These windings therefore tend to dampen out the load or other transients on the machine. It is for this reason that amortisseur windings are also called *damper windings*. Amortisseur windings are also used on synchronous generators,

# SG AND SM (6.4)

1. The distinguishing characteristic of a synchronous generator (supplying  $P$ ) is that  $E_A$  lies ahead of  $V_\phi$  while for a motor  $E_A$  lies behind  $V_\phi$ .
2. The distinguishing characteristic of a machine supplying reactive power  $Q$  is that  $E_A \cos \delta > V_\phi$  regardless of whether the machine is acting as a generator or as a motor. A machine that is consuming reactive power  $Q$  has  $E_A \cos \delta < V_\phi$ .



|                                                               | Supply reactive power $Q$<br>$E_A \cos \delta > V_\phi$ | Consume reactive power $Q$<br>$E_A \cos \delta < V_\phi$ |
|---------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------|
| Supply power $P$<br><br>Generator<br><br>$E_A$ leads $V_\phi$ |                                                         |                                                          |
| Consume power $P$<br><br>Motor<br><br>$E_A$ lags $V_\phi$     |                                                         |                                                          |

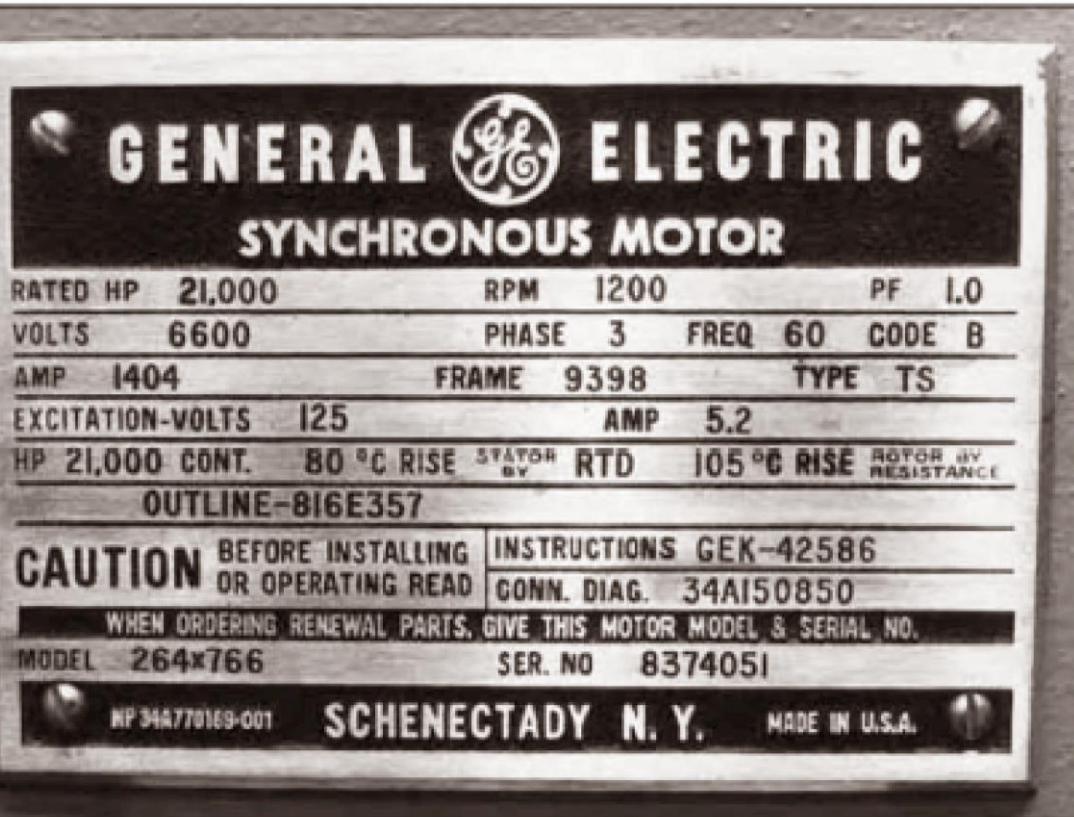
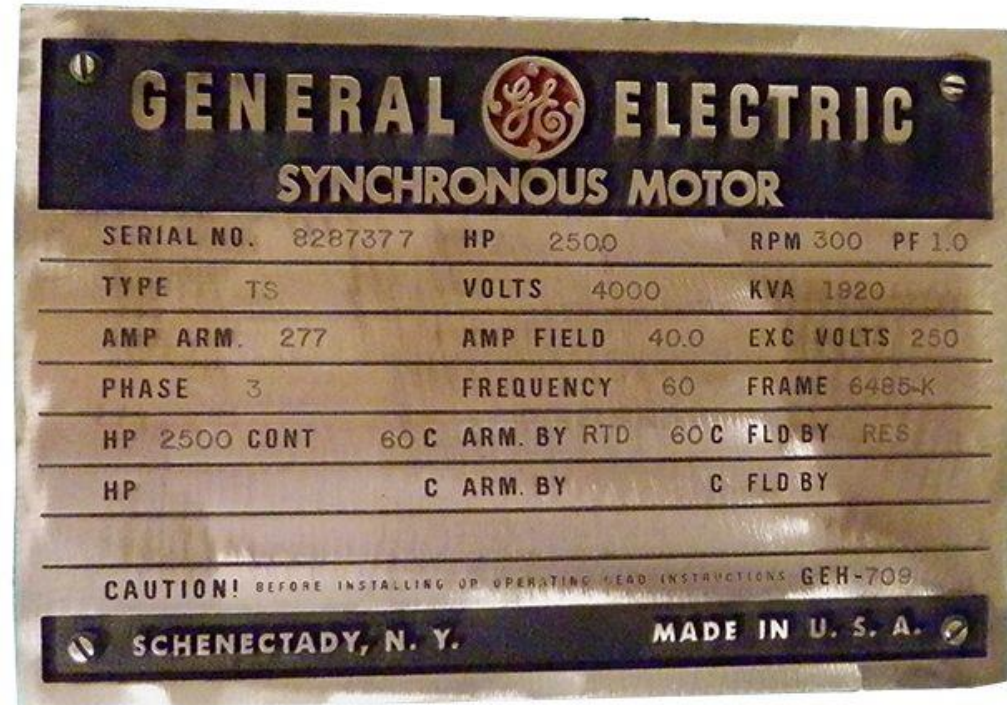


# SM RATINGS (6.5)

Ref [7]:  
<https://www.caseyusa.com/products/spec-sheet/109754000-00-2250-hp-900-rpm-2300-v-general-electric-8511y-ac-synchronous-motor>



Ref [8]:  
<https://www.pinterest.com/pin/516154807279184120/>



Ref [1]: Electric Machinery Fundamentals (4th Edition) - Stephen J. Chapman

# STOPPING SYNCHRONOUS MOTOR

- 1.The full DC excitation is maintained and the armature is short circuited.
- 2.The full DC excitation is maintained with armature connected to external resistors.
- 3.Mechanical braking may be applied.



# COMPARISON OF SM AND IM

| S. No. | Particular      | Synchronous Motor                                                             | 3-phase Induction Motor                |
|--------|-----------------|-------------------------------------------------------------------------------|----------------------------------------|
| 1.     | Speed           | Remains constant (i.e., $N_s$ ) from no-load to full-load.                    | Decreases with load.                   |
| 2.     | Power factor    | Can be made to operate from lagging to leading power factor.                  | Operates at lagging power factor.      |
| 3.     | Excitation      | Requires d.c. excitation at the rotor.                                        | No excitation for the rotor.           |
| 4.     | Economy         | Economical for speeds below 300 r.p.m.                                        | Economical for speeds above 600 r.p.m. |
| 5.     | Self-starting   | No self-starting torque.<br>Auxiliary means have to be provided for starting. | Self-starting                          |
| 6.     | Construction    | Complicated                                                                   | Simple                                 |
| 7.     | Starting torque | More                                                                          | less                                   |



# REFERENCES

Ref [1]: Electric Machinery Fundamentals (4th Edition) - Stephen J. Chapman

Ref [2]: Principles of Electrical Machines- V.K. Mehta and Rohit Mehta, 2nd Edition

Ref [3]: <https://www.pinterest.com/pin/402650022917293432/>

Ref [4]: <https://www.gepowerconversion.com/press-releases/largest-motor-ever-produced-peterborough-out-door>

Ref [5]: [https://www.industrial-electronics.com/Motor\\_Control\\_8-39.html](https://www.industrial-electronics.com/Motor_Control_8-39.html)

Ref [6]: <https://electrical-engineering-portal.com/synchronous-motor-construction>

Ref [7]: [https://www.caseyusa.com/products/spec\\_sheet/10975400000-2250-hp-900-rpm-2300-v-general-electric-8511y-ac-synchronous-motor](https://www.caseyusa.com/products/spec_sheet/10975400000-2250-hp-900-rpm-2300-v-general-electric-8511y-ac-synchronous-motor)

Ref [8]: <https://www.pinterest.com/pin/516154807279184120/>

## Youtube Links:

1. <https://www.youtube.com/watch?v=SiZ-mak4h4s>



*The End*

This image is a vintage-style title card. The central text, "The End", is written in a classic, elegant cursive script. The background of the card is a dark, textured surface that resembles a film strip, with vertical lines and some minor imperfections or dust specks. The card is set against a light gray background. In the corners of the overall image, there are several realistic-looking water droplets of various sizes, some of which are partially cut off by the edges. The overall aesthetic is nostalgic and cinematic.